CUC
City University of Cairo
جامعة المدينة بالقاهرة

MATHEMATICS ONE

Course Code: BAS111



Student Book

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Mathematics 1

(BAS111)

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Preface

The calculus course is a critical course for science, technology, engineering and math majors. This course sets the stage for many majors and is where students see the beauty of mathematics, encouraging them to take upper –level math courses.

In a calculus market-research study conducted in 2008, calculus faculty pointed out three critical components to student success in the calculus.

The most critical is mastery of the prerequisite algebra and trigonometry skills. Our market-research study showed that 58 percent mentioned that student struggled with calculus because of poor algebra skills and 72 percent said because of poor trigonometry skills. This is the number one learning challenge preventing student from being successful in the first calculus course.

The second critical component for student success is text that presents calculus concepts, especially the most challenging concepts, in a clear and elegant manner. This helps students see and appreciate the beauty and power of mathematics.

Lastly, calculus faculty told us that it is critical for a calculus text to include all the classic calculus problems.

Chapter (1) DIFFERENTIATION AND ITS APPLICATIONS

1.1. The inverse trigonometric functions

$$y = \sin^{-1} x$$

this means that

$$\sin y = x$$

Noting that

$$\sin\left(\sin^{-1} x\right) = x$$
$$\sin^{-1}\left(\sin x\right) = x$$

Similarly, for

$\cos^{-1} x$

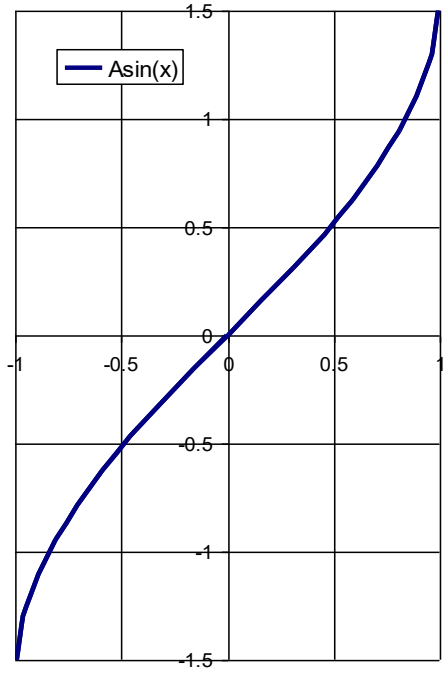
$\tan^{-1} x$

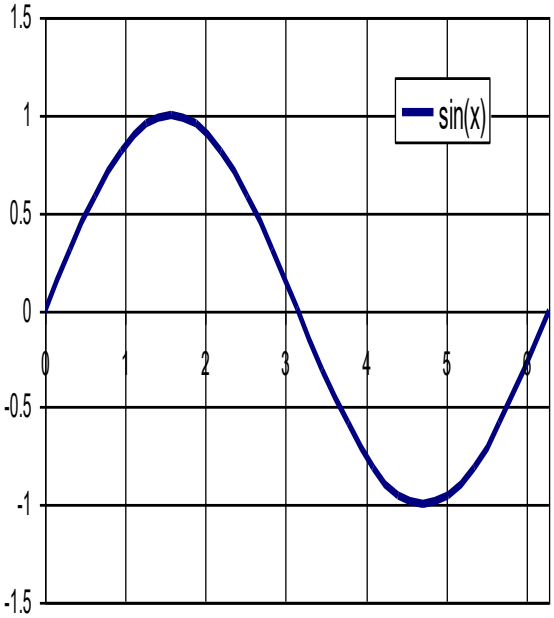
$\cot^{-1} x$

$\sec^{-1} x$

$\operatorname{cosec}^{-1} x$

Sin^{-1}x

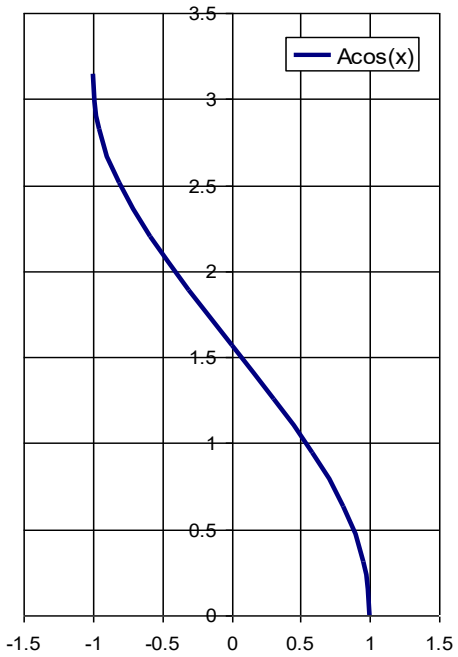


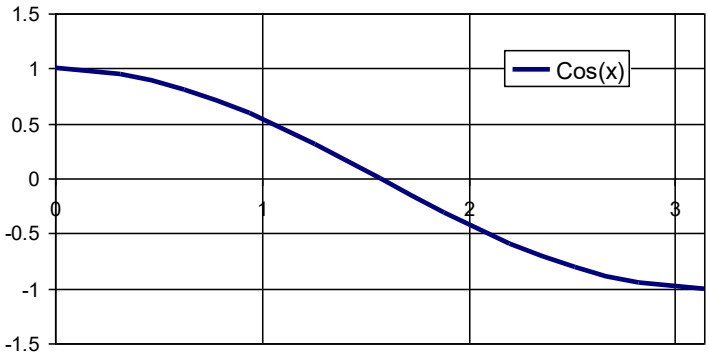


$$y = \sin^{-1}\frac{1}{2} \Rightarrow \sin y = \sin\left(\sin^{-1}\frac{1}{2}\right) = \frac{1}{2} \Rightarrow y = 30^\circ = \frac{\pi}{6}$$

You have to know how to use calculators!

$Cos^{-1}x$

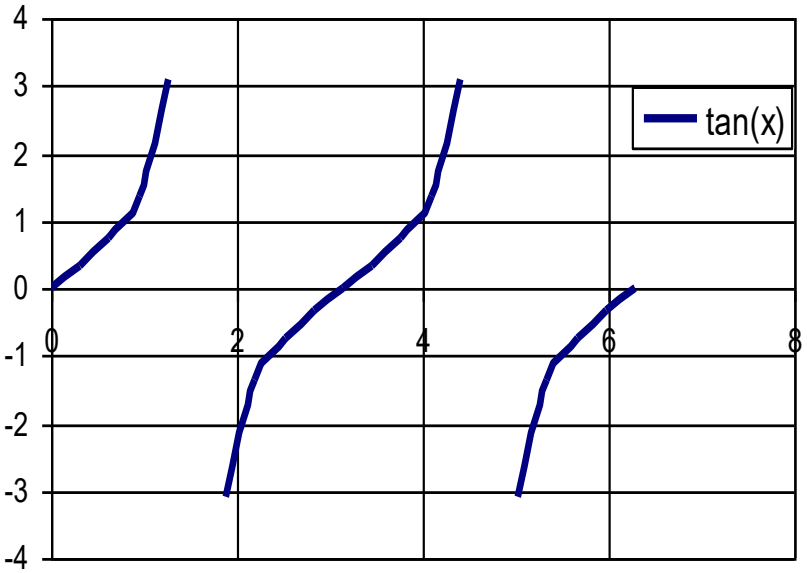


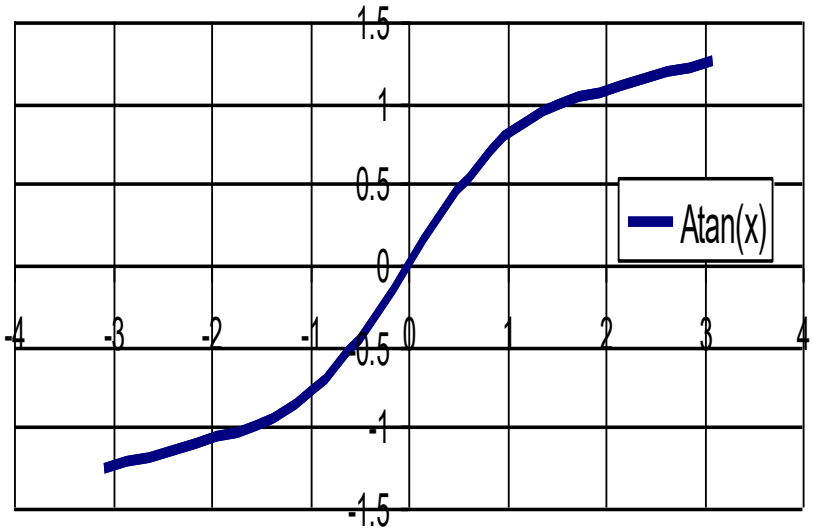


$$y = \cos^{-1}\frac{1}{2} \Rightarrow \cos y = \cos\left(\cos^{-1}\frac{1}{2}\right) = \frac{1}{2} \Rightarrow y = 60^\circ = \frac{\pi}{3}$$

You have to know how to use calculator to compute these functions!

$tan^{-1}x$





$$y = \tan^{-1}\sqrt{3} \Rightarrow \tan y = \tan\left(\tan^{-1}\sqrt{3}\right) = \sqrt{3} \Rightarrow y = 60^\circ = \frac{\pi}{3}$$

1.2. Derivatives of the inverse trigonometric functions

$$f(x) = \sin^{-1} x \longrightarrow f'(x) = \frac{1}{\sqrt{1-x^2}}$$

$$f(x) = \cos^{-1} x \longrightarrow f'(x) = \frac{-1}{\sqrt{1-x^2}}$$

$$f(x) = \tan^{-1} x \longrightarrow f'(x) = \frac{1}{1+x^2}$$

$$f(x) = \cot^{-1} x \longrightarrow f'(x) = \frac{-1}{1+x^2}$$

$$f(x) = \sec^{-1} x \longrightarrow f'(x) = \frac{1}{x\sqrt{x^2-1}}$$

$$f(x) = \operatorname{cosec}^{-1} x \longrightarrow f'(x) = \frac{-1}{x\sqrt{x^2-1}}$$

Examples

(1) $y = \sin^{-1}(3x^2 + 7)$

$$y' = \frac{1}{\sqrt{1-(3x^2+7)^2}} \times (6x)$$

(2) $y = \tan^{-1}(\cos x)$

$$y' = \frac{1}{1+(\cos x)^2} \times (-\sin x)$$

(3) $y = (2x^5 - 6)^4 \sec^{-1} x$

$$y' = 4(2x^5-6)^3(10x^4)\sec^{-1}x + (2x^5-6)^4 \frac{1}{x\sqrt{x^2-1}}$$

(4) $y = \dfrac{\sin^{-1} x}{x^3 - 1}$

$$y' = \frac{\frac{1}{\sqrt{1-x^2}}(x^3-1) - \sin^{-1}x\,(3x^2)}{(x^3-1)^2}$$

(5) Prove that the derivative of $y = \sin^{-1} x$ is

$$y' = \frac{1}{\sqrt{1-x^2}}$$

Solution

$$y = \sin^{-1} x \longrightarrow \sin y = x$$

By differentiating with respect to x

$$\cos y \; y' = 1 \longrightarrow y' = \frac{1}{\cos y}$$??

How to calculate $\cos y$

We have $\cos^2 y + \sin^2 y = 1$

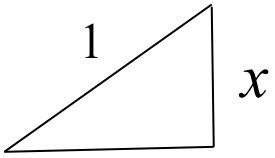
Then $\cos y = \sqrt{1-\sin^2 y}$ $\sin y = x$

$= \sqrt{1-x^2}$

Thus $y' = \frac{1}{\sqrt{1-x^2}}$

Another method to calculate $\cos y$

$\sin y = x = \frac{x}{1}$ ⟶



$\frac{y}{\sqrt{1-x^2}}$

Then

$\cos y = \frac{\sqrt{1-x^2}}{1} = \sqrt{1-x^2}$

Thus $y' = \frac{1}{\sqrt{1-x^2}}$

This is by using the Pythagoras theorem

1.3.Exponential and logarithmic Functions

- **Definition of e**

$$e = \lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n \approx 2.71828\cdots\cdots$$

$f(x) = e^x$ ⟶ $f'(x) = e^x$

Examples

(1) $y = e^{x^3+1}$

$$y' = e^{x^3+1} \times (3x^2)$$

(2) $y = e^{\sin^{-1} x}$

$$y' = e^{\sin^{-1} x} \frac{1}{\sqrt{1-x^2}}$$

- **The Logarithmic Function**

$$y = Log_a x \xrightarrow{\text{means}} a^y = x$$

For example If $y = Log_2 8$

Then we search for y such that $2^y = 8$

Then $y = 3$

- **Rules of logarithms for** $a > 1$

(1) $Log_a(xy) = Log_a x + Log_a y$

(2) $Log_a(x/y) = Log_a x - Log_a y$

(3) $Log_a x^n = n\, Log_a x$

(4) $Log_a a = 1$ ↓ $a^1 = a$

(5) $Log_a 1 = 0$ ↑ $a^0 = 1$

(6) $Log_a \infty = \infty$ ↓ $a^{\infty} = \infty$

(7) $Log_a 0 = -\infty$ ↓ $a^{-\infty} = \frac{1}{a^{\infty}} = \frac{1}{\infty} = 0$

$$\boxed{f(x) = Ln\,x} \qquad Ln\,x \equiv Log_e x$$

Then

$$y = Ln\,x \xrightarrow{\text{means}} e^y = x$$

For example

$$Ln\,1 = 0 \longleftrightarrow \boxed{e^0 = 1}$$

$$Ln\,0 = -\infty \longleftrightarrow \boxed{e^{-\infty} = 0}$$

- **Derivative of** $Ln\,x$

$$f(x) = Ln\,x \longrightarrow f\,'(x) = \frac{1}{x}$$

Examples

(1) $y = \ln(3x^2 + 5)$

$$y' = \frac{1}{3x^2 + 5} \times (6x)$$

(2) $y = \ln \tan x$

$$y' = \frac{1}{\tan x} \sec^2 x$$

Noting that

$$f(x) = Log_a x \longrightarrow f\,'(x) = \frac{1}{x}\,\frac{1}{\ln a}$$

$$f(x) = a^x \longrightarrow f\,'(x) = a^x\,Ln\,a$$

Can you explain why?

Examples

(3) $y = Log_5(x^3 + 8)$

$$y' = \frac{1}{x^3 + 8}(3x^2)\frac{1}{\ln 5}$$

(4) $y = 3^x$

$$y' = 3^x\,Ln\,3$$

- **Important Examples**

Example

If $y = x^x$ find y'

solution

$$y = x^x \xrightarrow{\text{Taking Ln both sides}} Ln\, y = Ln\, x^x = x\, Ln\, x$$

By differentiating both sides w.r.t. x

$$\frac{1}{y} y' = Ln\, x + x \frac{1}{x}$$

Then

$$y' = y\,(Ln\, x + 1) = x^x\,(Ln\, x + 1)$$

Example

If $y = (x^2 + 1)^{\sin x}$ find y'

solution

$$Ln\, y = Ln\,(x^2 + 1)^{\sin x} = \sin x\; Ln\,(x^2 + 1)$$

By differentiating both sides w.r.t. x

$$\frac{1}{y} y' = \cos x\; Ln(x^2 + 1) + \sin x \frac{1}{x^2 + 1}(2x)$$

Then

$$y' = y\left(\cos x\; Ln(x^2 + 1) + \sin x \frac{1}{x^2 + 1}(2x)\right)$$

Example

If $y = \dfrac{(x^2+1)^7(4x^3-5)^2}{(x+4)^3}$ find y'

solution

$$Ln\, y = 7Ln(x^2+1) + 2Ln(4x^3-5) - 3Ln(x+4)$$

By differentiating both sides w.r.t.

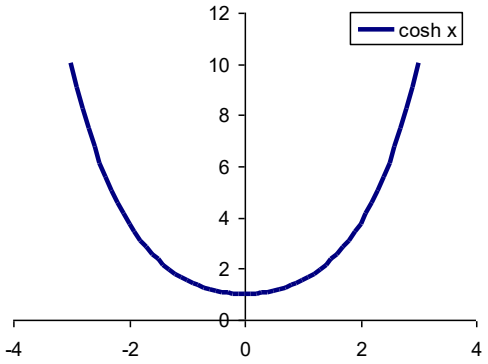
$$\frac{1}{y}y' = 7\frac{1}{x^2+1}(2x) + 2\frac{1}{4x^3-5}(12x^2) - 3\frac{1}{x+4}$$

Then

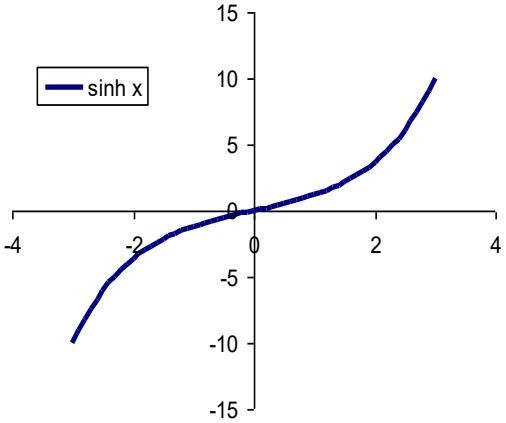
$$y' = y\left(7\frac{1}{x^2+1}(2x) + 2\frac{1}{4x^3-5}(12x^2) - 3\frac{1}{x+4}\right)$$

1.4. The Hyperbolic functions

$$\cosh x = \frac{e^x + e^{-x}}{2}$$



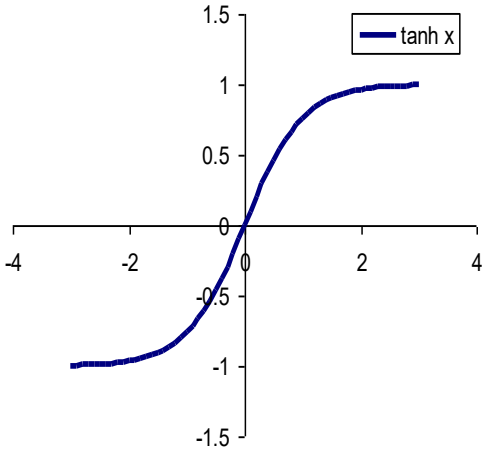
$$\sinh x = \frac{e^x - e^{-x}}{2}$$

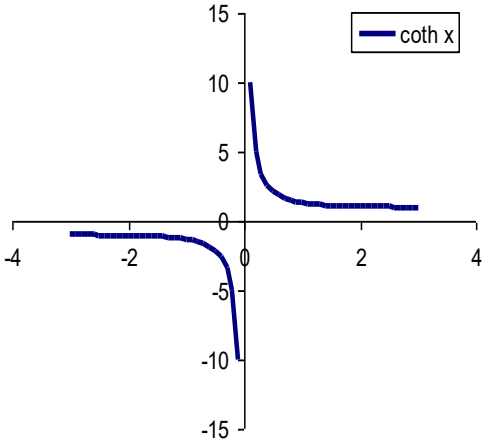


Try to use the calculator
to calculate these functions

$$\tanh x = \frac{\sinh x}{\cosh x} = \frac{e^{x} - e^{-x}}{e^{x} + e^{-x}}$$

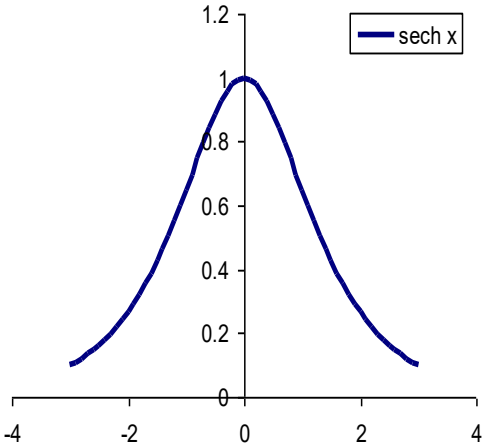
$$\coth x = \frac{\cosh x}{\sinh x} = \frac{e^{x} + e^{-x}}{e^{x} - e^{-x}}$$

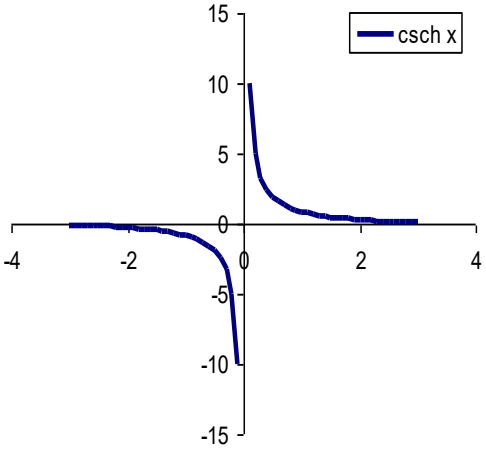




$$\operatorname{sech} x = \frac{1}{\cosh x} = \frac{2}{e^{x} + e^{-x}}$$

$$\operatorname{cosech} x = \frac{1}{\sinh x} = \frac{2}{e^{x} - e^{-x}}$$





Try to prove that

$$\cosh^2 x - \sinh^2 x = 1 \quad ??$$

Hint

Use that

$$\cosh x = \frac{e^{x} + e^{-x}}{2}$$

$$\sinh x = \frac{e^{x} - e^{-x}}{2}$$

- **How to differentiate these functions?**

$$y = \cosh x = \frac{e^x + e^{-x}}{2}$$

Then

$$y' = \frac{e^x + e^{-x}(-1)}{2} = \frac{e^x - e^{-x}}{2} = \sinh x$$

Similarly, for the rest of these functions

- **Derivative of Hyperbolic functions**

$$f(x) = \sinh x \longrightarrow f'(x) = \cosh x$$
$$f(x) = \cosh x \longrightarrow f'(x) = \sinh x$$
$$f(x) = \tanh x \longrightarrow f'(x) = sech^2 x$$
$$f(x) = \coth x \longrightarrow f'(x) = -\text{cosech}^2 x$$
$$f(x) = sechx \longrightarrow f'(x) = -sechx \tanh x$$
$$f(x) = \text{cosech}x \longrightarrow f'(x) = -\text{cosech}x \coth x$$

Examples

(8) $y = (x^2 + 1)\cosh x$

$$y' = 2x \cosh x + (x^2 + 1)\sinh x$$

(9) $y = \dfrac{\tanh(6x + 1)}{x^4 - 5}$

$$y' = \frac{sech^2(6x + 1)(6)(x^4 - 5) - \tanh(6x + 1)(4x^3)}{(x^4 - 5)^2}$$

1.5. The inverse hyperbolic functions

$y = \sinh^{-1} x$ this means that

$$\sinh y = x$$

Similarly, for

$\cosh^{-1} x$

$\tanh^{-1} x$

$\coth^{-1} x$

$\operatorname{sech}^{-1} x$

$\operatorname{cosech}^{-1} x$

Noting that

$$\sinh\left(\sinh^{-1} x\right) = x$$
$$\sinh^{-1}\left(\sinh x\right) = x$$

- **Derivatives of the inverse hyperbolic functions**

$$f(x) = \sinh^{-1} x \longrightarrow f'(x) = \frac{1}{\sqrt{x^2+1}}$$

$$f(x) = \cosh^{-1} x \longrightarrow f'(x) = \frac{1}{\sqrt{x^2-1}}$$

$$f(x) = \tanh^{-1} x \longrightarrow f'(x) = \frac{1}{1-x^2}$$

$$f(x) = \coth^{-1} x \longrightarrow f'(x) = \frac{-1}{x^2-1}$$

$$f(x) = \operatorname{sech}^{-1} x \longrightarrow f'(x) = \frac{-1}{x\sqrt{1-x^2}}$$

$$f(x) = \operatorname{cosech}^{-1} x \longrightarrow f'(x) = \frac{-1}{x\sqrt{1+x^2}}$$

Examples

(10) $y = \sinh^{-1}(x^4 + 7)$

$$y' = \frac{1}{\sqrt{(x^4+7)^2 + 1}} \times (4x^3)$$

(11) $y = \tanh^{-1}(Ln\, x)$

$$y' = \frac{1}{1-(Ln\, x)^2} \times (\frac{1}{x})$$

Example

Prove that the derivative of $y = \sinh^{-1} x$ is $y' = \frac{1}{\sqrt{x^2+1}}$

Solution

$y = \sinh^{-1} x \longrightarrow \sinh y = x$

By differentiating with respect to x

$\cosh y \; y' = 1 \longrightarrow y' = \frac{1}{\cosh y}$??

How to calculate $\cosh y$

We have $\cosh^2 y - \sinh^2 y = 1$

Then $\cosh y = \sqrt{\sinh^2 y + 1}$ $\sinh y = x$

$= \sqrt{x^2+1}$

Thus $y' = \frac{1}{\sqrt{x^2+1}}$

Important Example

Example

Prove that $\sinh^{-1} x = \ln\left(x + \sqrt{x^2+1}\right)$

Solution

Let $y = \sinh^{-1} x \longrightarrow \sinh y = x$

$$\frac{e^y - e^{-y}}{2} = x \quad \times 2$$

$$e^y - e^{-y} = 2x$$

let

$e^y = z \longrightarrow e^{-y} = \frac{1}{e^y} = \frac{1}{z}$

Then $z - \frac{1}{z} = 2x \quad \times z$

$$z^2 - 1 = 2xz$$

$$z^2 - 2xz - 1 = 0$$

$a = 1 \ , \ b = -2x \ , \ c = -1$

$$az^2 + bz + c = 0$$

$$z = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$z = \frac{2x \pm \sqrt{4x^2+4}}{2} = \frac{2x \pm 2\sqrt{x^2+1}}{2} = x \pm \sqrt{x^2+1}$$

but $e^y = z$

$$e^y = x \pm \sqrt{x^2+1}$$

taking Ln both sides and considering only the positive sign

$$y = Ln\left(x + \sqrt{x^2+1}\right) \qquad \boxed{y = \sinh^{-1} x}$$

$$\sinh^{-1} x = Ln\left(x + \sqrt{x^2+1}\right)$$

Try to use this result
to calculate the derivative??

1.6. Applications of The Derivative

- **The Nth Derivative of a Function**

We know that the first derivative is denoted by

$$\frac{dy}{dx} = y' \quad \text{or} \quad \frac{dy}{dx} = y^{(1)}$$

For derivatives of higher order, we are going to use the following notations

$$\frac{d^2y}{dx^2} = y'' = y^{(2)} \;,\; \frac{d^3y}{dx^3} = y''' = y^{(3)} \;,\; \frac{d^4y}{dx^4} = y^{(4)} \;,\cdots$$

Generally, the n[th] derivative will be denoted $\boxed{\frac{d^n y}{dx^n} = y^{(n)}}$

Example

Calculate the n[th] derivative of the function $y = e^{3x}$

Solution

$$y^{(1)} = 3e^{3x}$$

We have

$$y^{(2)} = 3^2 e^{3x}$$

$$y^{(3)} = 3^3 e^{3x}$$

Generally, the n[th] derivative will be $\boxed{y^{(n)} = 3^n e^{3x}}$

- **Important Trigonometric Rules**

(1) $-\sin(\theta) = \cos(\theta + \frac{\pi}{2})$

(2) $\cos(\theta) = \sin(\theta + \frac{\pi}{2})$

To prove (1), we have

$$(1)\ R.H.S. = \cos(\theta + \frac{\pi}{2}) = \cos\theta\cos\frac{\pi}{2} - \sin\theta\sin\frac{\pi}{2}$$
$$= \cos\theta\ (0) - \sin\theta\ (1) = -\sin\theta$$

To prove (2), we have

$$(2)\ R.H.S. = \sin(\theta + \frac{\pi}{2}) = \sin\theta\cos\frac{\pi}{2} + \cos\theta\sin\frac{\pi}{2}$$
$$= \sin\theta\ (0) + \cos\theta\ (1) = \cos\theta$$

Example

Calculate the n[th] derivative of the function $y = \cos(4x)$

Solution

This is by using the previous rules

We have $y^{(1)} = -4\sin(4x) = 4\cos(4x + \frac{\pi}{2})$

$$y^{(2)} = -4^2 \sin(4x + \frac{\pi}{2}) = 4^2 \cos(4x + 2\frac{\pi}{2})$$

$$y^{(3)} = -4^3 \sin(4x + 2\frac{\pi}{2}) = 4^3 \cos(4x + 3\frac{\pi}{2})$$

Generally, the n[th] derivative will be $\boxed{y^{(n)} = 4^n \cos(4x + n\frac{\pi}{2})}$

Example

Calculate the n[th] derivative of the function $y = \sin(6x+5)$

Solution

By using similar steps to those in example 2, the n[th] derivative will be on the form

$$y^{(n)} = 6^n \sin(6x+5+n\frac{\pi}{2})$$

Try to prove this by yourself

Example

Calculate the n[th] derivative of the $y = \frac{1}{5x+7}$

Solution

We have $y = \frac{1}{5x+7} = (5x+7)^{-1}$

Then $y^{(1)} = (-1)(5x+7)^{-2}5 = (-1)(1)(5x+7)^{-2}5$

$y^{(2)} = (-1)(-2)(5x+7)^{-3}5^2 = (-1)^2(1\times 2)(5x+7)^{-3}5^2$

$y^{(3)} = (-1)(-2)(-3)(5x+7)^{-4}5^3 = (-1)^3(1\times 2\times 3)(5x+7)^{-4}5^3$

Generally, we can write

$$y^{(n)} = (-1)^n(1\times 2\times 3\times\cdots\times n)(5x+7)^{-(n+1)}5^n$$

$$= (-1)^n n!(5x+7)^{-(n+1)}5^n$$

The n[th] derivative will be

$$y^{(n)} = \frac{(-1)^n n!\,5^n}{(5x+7)^{n+1}}$$

Some general rules

y	$y^{(n)}$
e^{ax+b}	$a^n e^{ax+b}$
$\cos(ax+b)$	$a^n \cos(ax+b+n\frac{\pi}{2})$
$\sin(ax+b)$	$a^n \sin(ax+b+n\frac{\pi}{2})$
$\frac{1}{ax+b}$	$\frac{(-1)^n a^n n!}{(ax+b)^{n+1}}$

Leibniz Rule

Leibniz rule is a rule that gives the nth derivative of the product of two functions. It is on the form

$$(fg)^{(n)} = f^{(n)}g^{(0)} + \frac{n}{1!}f^{(n-1)}g^{(1)} + \frac{n(n-1)}{2!}f^{(n-2)}g^{(2)} + \cdots + f^{(0)}g^{(n)}$$

General ideas about applying Leibniz rule

(1) Always choose g to be the function that its derivative is zero after finite number of differentiations

(2) Always choose f to be the function that it is easy to find its nth derivative

Example

Calculate the nth derivative of the function $y = (3x^2+1)e^{5x}$

Solution

Leibniz rule is on the form

$$(fg)^{(n)} = f^{(n)}g^{(0)} + \frac{n}{1!}f^{(n-1)}g^{(1)} + \frac{n(n-1)}{2!}f^{(n-2)}g^{(2)} + \cdots + f^{(0)}g^{(n)}$$

→ Equation (1)

By choosing $g = (3x^2+1)$, $f = e^{5x}$ then we have the following

$$g^{(1)} = 6x$$

$$g^{(2)} = 6$$

$$g^{(3)} = g^{(4)} = g^{(5)} = \cdots = 0$$

Then equation (1) will take the form

$$(fg)^{(n)} = f^{(n)}(3x^2+1) + \frac{n}{1!}f^{(n-1)}(6x) + \frac{n(n-1)}{2!}f^{(n-2)}(6) + 0 + \cdots + 0$$

→ Equation (2)

Now we have

$$f = e^{5x}$$

$$f^{(n)} = 5^n e^{5x}$$

$$f^{(n-1)} = 5^{n-1}e^{5x} \quad , \quad f^{(n-2)} = 5^{n-2}e^{5x}$$

Then equation (2) will take the form

$$(fg)^{(n)} = 5^n e^{5x}(3x^2+1) + \frac{n}{1!}5^{n-1}e^{5x}(6x) + \frac{n(n-1)}{2!}5^{n-2}e^{5x}(6)$$

Now by simplifying the last equation we can write

$$\left((3x^2+1)e^{5x}\right)^{(n)} = 5^n(3x^2+1)e^{5x} + 6nx5^{n-1}e^{5x} + 3n(n-1)5^{n-2}e^{5x}$$

Example

Calculate the nth derivative of the $y = 4x\sin(3x)$

Solution

Leibniz rule is on the

$$(fg)^{(n)} = f^{(n)}g^{(0)} + \frac{n}{1!}f^{(n-1)}g^{(1)} + \frac{n(n-1)}{2!}f^{(n-2)}g^{(2)} + \cdots + f^{(0)}g^{(n)}$$

→ Equation (1)

By choosing $g = 4x$, $f = \sin(3x)$ then we have the

$$g^{(1)} = 4$$
$$g^{(2)} = g^{(3)} = g^{(4)} = \cdots = 0$$

Then equation (1) will take the form

$$(fg)^{(n)} = f^{(n)}(4x) + \frac{n}{1!} f^{(n-1)}(4) + 0 + \cdots + 0$$

→ Equation (2)

Now we have

$$f = \sin(3x)$$
$$f^{(n)} = 3^n \sin(3x + n\frac{\pi}{2}) \quad , \quad f^{(n-1)} = 3^{n-1} \sin(3x + (n-1)\frac{\pi}{2})$$

Then equation (2) will take the form

$$(fg)^{(n)} = 3^n \sin(3x + n\frac{\pi}{2})(4x) + \frac{n}{1!} 3^{n-1} \sin(3x + (n-1)\frac{\pi}{2})(4)$$

Now by simplifying the last equation we can write

$$\left(4x\sin(3x)\right)^{(n)} = 4x3^n \sin(3x + n\frac{\pi}{2}) + 4n\ 3^{n-1} \sin(3x + (n-1)\frac{\pi}{2})$$

1.7.L'Hopital Rule

If *f(x)* and *g(x)* are such that

$$\lim_{x\to a} \frac{f(x)}{g(x)} = \frac{0}{0} \qquad \frac{\infty}{\infty}$$

Then we have

$$\lim_{x\to a} \frac{f(x)}{g(x)} = \lim_{x\to a} \frac{f'(x)}{g'(x)}$$



The French Mathematician Guillaume de *L'Hopital*

Example

By using L'Hopital $\lim_{x\to 1}\frac{Ln\,x}{x-1}$

Solution

We can apply L'Hopital rule

$$\lim_{x\to 1}\frac{Ln\,x}{x-1}=\frac{Ln\,1}{1-1}=\frac{0}{0}$$

Then

$$\lim_{x\to 1}\frac{Ln\,x}{x-1}=\lim_{x\to 1}\frac{1/x}{1}=1$$

Example

By using L'Hopital rule $\lim_{x\to\infty}\frac{e^{3x}}{x^2}$

Solution

We can apply L'Hopital

$$\lim_{x\to\infty}\frac{e^{3x}}{x^2}=\frac{\infty}{\infty}$$

We can apply L'Hopital rule

Then

$$\lim_{x\to\infty}\frac{e^{3x}}{x^2}=\lim_{x\to\infty}\frac{3e^{3x}}{2x}=\frac{\infty}{\infty}$$

Then

$$\lim_{x\to\infty}\frac{3e^{3x}}{2x}=\lim_{x\to\infty}\frac{9e^{3x}}{2}=\frac{\infty}{2}=\infty$$

Example

By using L'Hopital rule calculate $\lim_{x\to\infty}xe^{-4x}$

Solution

$\lim_{x\to\infty}xe^{-4x}=\infty\times 0$ We can't apply L'Hopital rule

But $\lim_{x\to\infty}xe^{-4x}=\lim_{x\to\infty}\frac{x}{e^{4x}}=\frac{\infty}{\infty}$

We can apply L'Hopital rule

Then

$$\lim_{x\to\infty}\frac{x}{e^{4x}}=\lim_{x\to\infty}\frac{1}{4e^{4x}}=\frac{1}{\infty}=0$$

Example

By using L'Hopital rule $\lim_{x\to 0^+} x\,Ln\,x$

Solution

We can't apply L'Hopital

$$\lim_{x\to 0^+} x\,Ln\,x = 0\times -\infty$$

We can apply L'Hopital

But

$$\lim_{x\to 0^+} x\,Ln\,x = \lim_{x\to 0^+} \frac{Ln\,x}{1/x} = \frac{-\infty}{\infty}$$

Then

$$\lim_{x\to 0^+} \frac{Ln\,x}{1/x} = \lim_{x\to 0^+} \frac{1/x}{-1/x^2} = \lim_{x\to 0^+} (-x) = 0$$

Example

By using L'Hopital rule calculate $\lim_{x\to \frac{\pi}{2}^+} (\tan x - \sec x)$

Solution

We can't apply L'Hopital rule

$$\lim_{x\to \frac{\pi}{2}^+} (\tan x - \sec x) = \infty - \infty$$

Bu

$$\lim_{x\to \frac{\pi}{2}^+} (\tan x - \sec x) = \lim_{x\to \frac{\pi}{2}^+} \left(\frac{\sin x}{\cos x} - \frac{1}{\cos x}\right) = \lim_{x\to \frac{\pi}{2}^+} \left(\frac{\sin x - 1}{\cos x}\right) = \frac{0}{0}$$

We can apply L'Hopital rule

Then

$$\lim_{x\to \frac{\pi}{2}^+} \left(\frac{\sin x - 1}{\cos x}\right) = \lim_{x\to \frac{\pi}{2}^+} \left(\frac{\cos x}{-\sin x}\right) = \frac{0}{-1} = 0$$

Example

By using L'Hopital rule calculate $\lim_{x\to 0^+}(1+3x)^{1/2x}$

Solution

$$\lim_{x\to 0^+}(1+3x)^{1/2x} = 1^{\infty}$$ We can't apply L'Hopital rule

Let $$A = \lim_{x\to 0^+}(1+3x)^{1/2x}$$

Then $$Ln\,A = \lim_{x\to 0^+} Ln(1+3x)^{1/2x}$$

Hence

$$Ln\,A = \lim_{x\to 0^+}\frac{1}{2x}Ln(1+3x) = \lim_{x\to 0^+}\frac{Ln(1+3x)}{2x} = \frac{0}{0}$$

We can apply L'Hopital rule

Then $$Ln\,A = \lim_{x\to 0^+}\frac{\left(\frac{1}{1+3x}\right)3}{2} = \frac{3}{2} \longrightarrow A = e^{3/2}$$

Thus $$A = \lim_{x\to 0^+}(1+3x)^{1/2x} = e^{3/2}$$

1.8. Maclaurin & Taylor Series

A function *f* has a power series (polynomial)

Maclaurin series

$$f(x) = \sum_{n=0}^{\infty}\frac{f^{(n)}(0)}{n!}x^n = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \ldots + \frac{f^{(n)}(0)}{n!}x^n + \ldots$$

Taylor series

$$f(x) = \sum_{n=0}^{\infty}\frac{f^{(n)}(c)}{n!}(x-c)^n = f(c) + \frac{f'(c)}{1!}(x-c) + \frac{f''(c)}{2!}(x-c)^2 + \ldots + \frac{f^{(n)}(c)}{n!}(x-c)^n + \ldots$$

Example

Find the Maclaurin series for e^x

Solution

$$f(x)=f(0)+\frac{f'(0)}{1!}x+\frac{f''(0)}{2!}x^2+\frac{f'''(0)}{3!}x^3+....$$

$$\begin{aligned} f(x)=e^x &\Rightarrow f(0)=1\\ f'(x)=e^x &\Rightarrow f'(0)=1\\ f''(x)=e^x &\Rightarrow f''(0)=1 \end{aligned}$$

$$e^x=1+x+\frac{1}{2!}x^2+\frac{1}{3!}x^3+\frac{1}{4!}x^4+...$$

Example

Find the Taylor series for ln$x$ at $x=1$

Solution

$$f(x)=f(1)+\frac{f'(1)}{1!}(x-1)+\frac{f''(1)}{2!}(x-1)^2+\frac{f'''(1)}{3!}(x-1)^3+....$$

$$\begin{aligned} f(x) &= \ln x && \Rightarrow f(1)=0\\ f'(x) &= \frac{1}{x} && \Rightarrow f'(1)=1\\ f''(x) &= \frac{-1}{x^2} && \Rightarrow f''(1)=-1\\ f'''(x) &= \frac{2}{x^3} && \Rightarrow f'''(1)=2 \end{aligned}$$

Then

$$\ln x=0+\frac{1}{1}(x-1)+\frac{-1}{2}(x-1)^2+\frac{2}{6}(x-1)^3+.....$$

$$\ln x=(x-1)-\frac{1}{2}(x-1)^2+\frac{1}{3}(x-1)^3+.....$$

Important Note

We can calculate an approximation for ln(1.2) as follows

From the last expansion

$$\ln x = (x-1) - \frac{1}{2}(x-1)^2 + \frac{1}{3}(x-1)^3 + ...$$

Then

$$\ln(1.2) = (1.2-1) - \frac{1}{2}(1.2-1)^2 + \frac{1}{3}(1.2-1)^3 + ...$$

Approximately we get

$$\ln(1.2) \approx (0.2) - \frac{1}{2}(0.2)^2 + \frac{1}{3}(0.2)^3 = 0.18266$$

Example

Find the series for $\mathrm{Sin}(x)$ at $x = 0$

and use this to approximate $\sin(\frac{2\pi}{7})$

Solution

$$f(x) = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + ...$$

$$\begin{array}{llll} f(x) & = \sin x & \Rightarrow f(0) & = 0 \\ f'(x) & = \cos x & \Rightarrow f'(0) & = 1 \\ f''(x) & = -\sin x & \Rightarrow f''(0) & = 0 \\ f^{(3)}(x) & = -\cos x & \Rightarrow f^{(3)}(0) & = -1 \\ f^{(4)}(x) & = \sin x & \Rightarrow f^{(4)}(0) & = 0 \end{array}$$

Then

$$\sin x = 0 + \frac{1}{1!}x + \frac{0}{2!}x^2 + \frac{-1}{3!}x^3 + \frac{0}{4!}x^4 + \frac{1}{5!}x^5 + ...$$

$$= x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - \frac{1}{7!}x^7 + ...$$

Now

$$\pi \approx 3.14$$

$$\frac{2\pi}{7} \approx 0.897$$

$$\pi^{rad} \equiv 180^\circ$$

Then

$$\sin(\frac{2\pi}{7}) = \sin(0.897) \approx 0.897 - \frac{1}{3!}(0.897)^3 + \frac{1}{5!}(0.897)^5 - \frac{1}{7!}(0.897)^7$$

$$\approx 0.78145$$

Example

Find an approximation for $\int_0^1 \sin(x^2)dx$

Solution

We have

$$\sin x \approx x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - \frac{1}{7!}x^7$$

Replace each x by x^2

Then

$$\sin(x^2) \approx x^2 - \frac{1}{3!}x^6 + \frac{1}{5!}x^{10} - \frac{1}{7!}x^{14}$$

$$\int_0^1 \sin(x^2)dx \approx \int_0^1 \left(x^2 - \frac{1}{3!}x^6 + \frac{1}{5!}x^{10} - \frac{1}{7!}x^{14} \right) dx$$

$$= \left[\frac{x^3}{3} - \frac{1}{3!}\frac{x^7}{7} + \frac{1}{5!}\frac{x^{11}}{11} - \frac{1}{7!}\frac{x^{15}}{15} \right]_{x=0}^{x=1}$$

$$= \left[\frac{1}{3} - \frac{1}{3!}\frac{1}{7} + \frac{1}{5!}\frac{1}{11} - \frac{1}{7!}\frac{1}{15} \right] - 0$$

$$= 0.3102681$$

Sheet [1]

Problem 1

Find the first derivative y' of the following (with respect to x)

(a) $x^2 + \tan^{-1} y = y^2 + 4x + 8$

(b) $y^3 + e^{x^2+1} = 3e^y + 7$

(c) $x^3 y^5 + \sec^{-1} y + e^{\sin x} - 7 = 0$

Problem 2

By using the chain rule calculate $\frac{dy}{dx}$ for the following

(a) $y = \cos^{-1}(t+1)$, $x = t^2 + 3t + 4$

(b) $y = e^{t^2} + t - 1$, $x = \sin(3t + 4)$

Problem 3

Prove the following

(a) $\frac{d}{dx}\left(\cos^{-1} x\right) = \frac{-1}{\sqrt{1-x^2}}$

(b) $\frac{d}{dx}\left(\tan^{-1} x\right) = \frac{1}{1+x^2}$

(c) $\frac{d}{dx}\left(\sec^{-1} x\right) = \frac{1}{x\sqrt{x^2-1}}$

Problem 4

Find the first derivative of the following functions

(a) $f(x) = Ln(\sin^{-1} x)$

(b) $f(x) = 3^{x^2+1}(4x^5 - 6)$

(c) $f(x) = (1+x^3)\cosh(\ln x)$

(d) $f(x) = \frac{\tanh(3x+1)}{x^5 - 1}$

(e) $f(x) = Ln(Ln\, x)$

(f) $f(x) = \frac{1 + Ln\, x}{\sinh(2x-1)}$

Problem 5

Find the first derivative y' of following by using the logarithmic differentiation.

(a) $y = x^{\sin^{-1} x}$

(b) $y = (1+x^2)^{\sec x}$

(c) $y = \dfrac{(x+1)^3 (x^2+8x-3)^5}{(x-1)^7 (x^3+2)^4}$

(d) $y = \dfrac{(x^2+1)^4 \sinh x}{(x^5+3)^8}$

Problem 6

Prove that:

(a) $\cosh^2 x - \sinh^2 x = 1$

(b) $1 - \tanh^2 x = \operatorname{sech}^2 x$

Problem 7

Find the first derivative of the following functions

(a) $y = \cosh^{-1}(x^4+1)$

(b) $y = e^{4x+1} \tanh^{-1}(x+2)$

(c) $y = \sinh^{-1}(\sin x)$

(d) $y = (3x^2 - 4x + 6)\operatorname{sech}^{-1} x$

Problem 8

Prove the following

(a) $\dfrac{d}{dx}\left(\cosh^{-1} x\right) = \dfrac{1}{\sqrt{x^2-1}}$

(b) $\dfrac{d}{dx}\left(\tanh^{-1} x\right) = \dfrac{1}{1-x^2}$

Problem 9

Prove the following

(a) $\cosh^{-1} x = Ln(x + \sqrt{x^2 - 1}) \quad , \quad x \geq 1$

(b) $\tanh^{-1} x = \frac{1}{2} Ln\left(\frac{1+x}{1-x}\right) \quad , \quad |x| < 1$

(c) $\text{sech}^{-1}\ x = Ln\left(\frac{1+\sqrt{1-x^2}}{x}\right) \quad , \quad 0 < x \leq 1$

Problem 10

Find the nth derivative of the following functions

(a) $y = e^{4x+7}$ (b) $y = \sin(2x+5)$

(c) $y = \cos(5x)$ (d) $y = \frac{1}{7x+9}$

(e) $y = Ln(ax+b)$ (f) $y = \frac{1}{(ax+b)^3}$

Problem 11

By using Leibniz rule find the nth derivative of the following functions

(a) $y = 3xe^{7x}$ (b) $y = (x^3+1)e^{2x}$ (c) $y = (4x^2+2)\cos(3x)$

Problem 12

If $y = \frac{\sin^{-1} x}{\sqrt{1-x^2}}$ prove that, $(1-x^2)y' - xy - 1 = 0$ and hence by applying Leibniz rule prove that,

$$(1-x^2)y^{(n+1)} - (2n+1)xy^{(n)} - n^2 y^{(n-1)} = 0$$

Problem 13

If $y=\left(\sinh^{-1}x\right)^2$ prove that, $(1+x^2)y''+xy'-2=0$ and hence by applying Leibniz rule prove that,

$$(1+x^2)y^{(n+2)}+(2n+1)xy^{(n+1)}+n^2y^{(n)}=0$$

Problem 14

By using L'Hopital rule calculate the following limits

(1) $\lim_{x\to 2}\frac{\ln(x-1)}{x-2}$

(2) $\lim_{x\to\infty}x^2e^{-x}$

(3) $\lim_{x\to 0}\frac{\sin 5x}{x}$

(4) $\lim_{x\to 0}\frac{\cos x+2x-1}{3x}$

(5) $\lim_{x\to 0}\frac{e^x+e^{-x}-2}{1-\cos 2x}$

(6) $\lim_{x\to 2}\frac{x^3-8}{x-2}$

(7) $\lim_{x\to 0^+}x\ln x$

(8) $\lim_{x\to\infty}\frac{\ln x}{\sqrt{x}}$

(9) $\lim_{x\to 0}\frac{x-\sin x}{x^3}$

(10) $\lim_{x\to 0}(1+5x)^{1/2x}$

(11) $\lim_{x\to 0^+}x^x$

(12) $\lim_{x\to 0^+}x^{\sin x}$

Problem 15

Find the Maclaurin expansion of the following functions (at $x=0$)

(1) $f(x)=e^{3x}$

(2) $f(x)=\frac{1}{1+3x}$

Problem 16

Find the Taylor expansion of $f(x)=\ln x$ at $x=1$ and use this expansion to approximate the value of $\ln(1.3)$

Problem 17

Find the Taylor expansion of $f(x)=\sin x$ at $x=0$ and use this expansion to approximate the value of $\int_0^1 \frac{\sin x}{x}dx$

Problem 18:

Show that the Taylor expansion of $f(x)=\cos x$ at $x=0$ is given by $\cos x = 1-\frac{x^2}{2!}+\frac{x^4}{4!}-\frac{x^6}{6!}+\ldots$ and use this expansion to approximate the value of

(i) $\cos(\frac{2\pi}{9})$

(ii) $\int_0^1 \cos(x^2)dx$

Chapter (2) TECHNIQUES OF INTEGRATIONS

2.1. Indefinite Integration

Indefinite integration means the anti-derivative

$$\int f(x)\,dx = g(x) + c \quad \text{Means} \quad \boxed{\frac{d}{dx}\big(g(x)\big) = f(x)}$$

Example 1

$$\int \frac{1}{1+x^2}\,dx = \tan^{-1}x + c \qquad \text{because} \qquad \frac{d}{dx}\left(\tan^{-1}x\right) = \frac{1}{1+x^2}$$

$$\int \frac{1}{x}\,dx = Ln\,x + c \qquad \text{because} \qquad \frac{d}{dx}(Ln\,x) = \frac{1}{x}$$

- **Rules of Integration**

Rule 1

If k is a constant $\int k\,dx = kx + c$

Rule 2

$$\int x^n\,dx = \frac{x^{n+1}}{n+1} + c \quad n \neq -1$$

In the case $n = -1$ $\int x^{-1}\,dx = \int \frac{1}{x}\,dx = Ln\,x + c$

Example 2

$$(a)\ \int x^3 + \sqrt{x} + \frac{1}{x^7} - 8\,dx = \int x^3 + x^{1/2} + x^{-7} - 8\,dx$$

$$= \frac{x^4}{4} + \frac{x^{3/2}}{3/2} + \frac{x^{-6}}{-6} - 8x + c$$

$$(b)\ \int x^{5/2} + \frac{1}{x} + 3\,dx = \frac{x^{7/2}}{7/2} + Ln\,x + 3x + c$$

$$(c)\ \int \left(x - \frac{1}{x}\right)^2 dx = \int x^2 - 2 + \frac{1}{x^2}\,dx = \frac{x^3}{3} - 2x + \frac{x^{-1}}{-1} + c$$

Rule 3

$$\int [f(x)]^n f'(x)\,dx = \frac{[f(x)]^{n+1}}{n+1} + c \qquad n \neq -1$$

Example 3

$$(a)\quad \int 6x(3x^2+7)^{10}\,dx = \frac{(3x^2+7)^{11}}{11} + c$$

$$(b)\ \int x(3x^2+7)^{10}\,dx \ \times\frac{6}{6}\ = \frac{1}{6}\int 6x(3x^2+7)^{10}\,dx = \frac{1}{6}\frac{(3x^2+7)^{11}}{11} + c$$

Rule 4

$$\int \frac{f'(x)}{f(x)}\,dx = Ln\,f(x) + c$$

Example 4

$$(a)\ \int \frac{6x}{3x^2+7}dx = Ln(3x^2+7) + c$$

$$(b)\ \int \frac{x}{3x^2+7}dx \ \times\frac{6}{6}\ = \frac{1}{6}\int \frac{6x}{3x^2+7}dx = \frac{1}{6}Ln(3x^2+7) + c$$

$$(c)\ \int \tan x\,dx = \int \frac{\sin x}{\cos x}\,dx = -\int \frac{-\sin x}{\cos x}\,dx$$

$$= -Ln(\cos x) + c$$

$$= Ln(\cos x)^{-1} + c = Ln(\sec x) + c$$

$$(d)\ \int \cot x\,dx = \int \frac{\cos x}{\sin x}\,dx = Ln(\sin x) + c$$

$$(e)\ \int \sec x\,dx \ \times\frac{\sec x+\tan x}{\sec x+\tan x}\ = \int \frac{\sec^2 x + \sec x\tan x}{\sec x+\tan x}\,dx$$

$$= Ln(\sec x + \tan x) + c$$

Rule 5

$$\int e^{f(x)} f'(x)\, dx = e^{f(x)} + c$$

Example 5

$$(a)\ \int e^{\tan x} \sec^2 x\, dx = e^{\tan x} + c$$

$$(b)\ \int 2x e^{x^2+1}\, dx = e^{x^2+1} + c$$

$$(c)\ \int x e^{x^2+1}\, dx \ \times\frac{2}{2} = \frac{1}{2}\int 2x e^{x^2+1}\, dx = \frac{1}{2} e^{x^2+1} + c$$

Rule 6

$$\int \sin f(x)\ \ f'(x)\, dx = -\cos f(x) + c$$

$$\int \cos f(x)\ \ f'(x)\, dx = \sin f(x) + c$$

Example 6

$$(a)\ \int 8x \sin(4x^2+1)\, dx = -\cos(4x^2+1) + c$$

$$(b)\ \int x \cos(3x^2+1)\, dx = \frac{1}{6}\sin(3x^2+1) + c$$

Example 7

$$\int \sin^7 x \cos x\, dx = \int (\sin x)^7 \cos x\, dx = \frac{\sin^8 x}{8} + c$$

Another solution

Let $u = \sin x \longrightarrow du = \cos x\, dx$

Then $\int \sin^7 x \cos x\, dx = \int u^7\, du = \frac{u^8}{8} + c = \frac{\sin^8 x}{8} + c$

Example 8

$$(a)\ \int \sin^2 x\,dx = \int \frac{1}{2}(1-\cos 2x)\,dx$$

$$= \frac{1}{2}\left(x - \frac{1}{2}\sin 2x\right) + c$$

$$(b)\ \int \cos^2 x\,dx = \int \frac{1}{2}(1+\cos 2x)\,dx$$

$$= \frac{1}{2}\left(x + \frac{1}{2}\sin 2x\right) + c$$

Example 9

$\int x\sqrt{x+1}\,dx$ Let $u^2 = x+1$ $\Rightarrow$ $u^2 - 1 = x$, $2u\,du = dx$

Then

$$\int x\sqrt{x+1}\,dx = \int (u^2-1)\sqrt{u^2}\ 2u\ du$$

$$= \int (u^2-1)2u^2\ du$$

$$= \int (2u^4 - 2u^2)\,du = 2\frac{u^5}{5} - 2\frac{u^3}{3} + c$$ But $u = (x+1)^{1/2}$

$$= \frac{2}{5}(x+1)^{5/2} - \frac{2}{3}(x+1)^{3/2} + c$$

Example 10

$$\int \frac{1}{9+x^2}\,dx = \int \frac{1}{9\left(1+\frac{x^2}{9}\right)}\,dx$$ Let $u = \frac{x}{3}$ $\longrightarrow$ $du = \frac{1}{3}dx$

Then $3\,du = dx$

$$\int \frac{1}{9+x^2}\,dx = \int \frac{1}{9(1+u^2)}\ 3\,du$$

$$= \frac{1}{3}\int \frac{1}{1+u^2}\,du = \frac{1}{3}\tan^{-1}u + c = \frac{1}{3}\tan^{-1}\left(\frac{x}{3}\right) + c$$

In general we have $\int \frac{1}{a^2+x^2}\,dx = \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right) + c$

Example 11

$$\int \frac{1}{\sqrt{16-x^2}}\,dx = \int \frac{1}{\sqrt{16\left(1-\frac{x^2}{16}\right)}}\,dx \qquad \text{Let} \quad u=\frac{x}{4} \longrightarrow du=\frac{1}{4}dx$$

$$4\,du = dx$$

Then

$$\int \frac{1}{\sqrt{16-x^2}}\,dx = \int \frac{1}{\sqrt{16\left(1-u^2\right)}}\,4\,du$$

$$= \int \frac{1}{\sqrt{1-u^2}}\,du = \sin^{-1}u + c = \sin^{-1}\left(\frac{x}{4}\right)+c$$

In general we have $$\int \frac{1}{\sqrt{a^2-x^2}}\,dx = \sin^{-1}\left(\frac{x}{a}\right)+c$$

- **Also we can prove the following**

$$\int \frac{1}{a^2-x^2}\,dx = \frac{1}{a}\tanh^{-1}\left(\frac{x}{a}\right)+c$$

$$\int \frac{1}{\sqrt{x^2+a^2}}\,dx = \sinh^{-1}\left(\frac{x}{a}\right)+c$$

$$\int \frac{1}{\sqrt{x^2-a^2}}\,dx = \cosh^{-1}\left(\frac{x}{a}\right)+c$$

Example 12

$$\int \frac{1}{25-(x+1)^2}\,dx = \int \frac{1}{25\left(1-\frac{(x+1)^2}{25}\right)}\,dx \qquad \text{Let} \quad u=\frac{x+1}{5}$$

$$\longrightarrow du=\frac{1}{5}dx$$

Then

$$5\,du = dx$$

$$\int \frac{1}{25-(x+1)^2}\,dx = \int \frac{1}{25\left(1-u^2\right)}\,5\,du$$

$$= \frac{1}{5}\int \frac{1}{1-u^2}\,du = \frac{1}{5}\tanh^{-1}u + c = \frac{1}{5}\tanh^{-1}\left(\frac{x+1}{5}\right)+c$$

Example 13

$$\int \frac{2x-1}{36+(x-3)^2}\,dx = \int \frac{2x-1}{36\left(1+\frac{(x-3)^2}{36}\right)}\,dx$$

Let $u = \frac{x-3}{6}$

$\rightarrow 6u+3 = x$

$\rightarrow 6\,du = dx$

Then

$$\int \frac{2x-1}{36+(x-3)^2}\,dx = \int \frac{2(6u+3)-1}{36(1+u^2)}\,6\,du = \int \frac{12u+5}{6(1+u^2)}\,du$$

$$= \int \frac{12u}{6(1+u^2)}\,du + \int \frac{5}{6(1+u^2)}\,du$$

Then

$$\int \frac{2x-1}{36+(x-3)^2}\,dx = \int \frac{2u}{(1+u^2)}\,du + \frac{5}{6}\int \frac{1}{(1+u^2)}\,du$$

$$= Ln(1+u^2) + \frac{5}{6}\tan^{-1}u + c$$

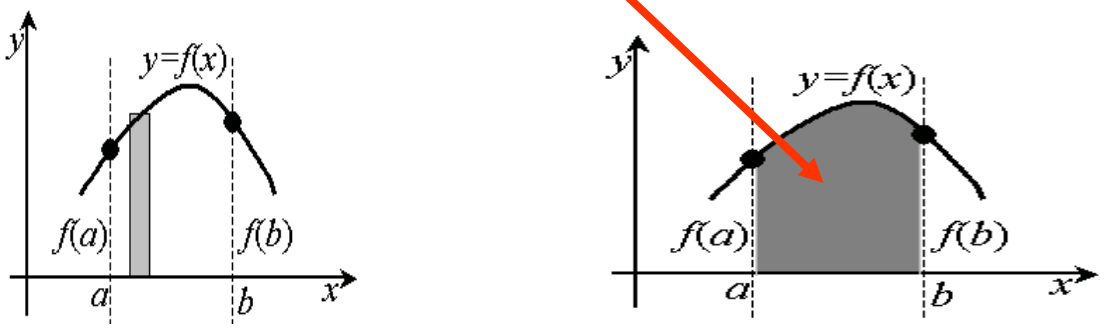
but $u = \frac{x-3}{6}$

$$\int \frac{2x-1}{36+(x-3)^2}\,dx = Ln\left[1+\left(\frac{x-3}{6}\right)^2\right] + \frac{5}{6}\tan^{-1}\left(\frac{x-3}{6}\right) + c$$

2.2. Definite Integrals

If f is integrable and $f(x) > 0$ for every x in $[a,b]$, then

$\int_a^b f(x)dx =$ area under the curve of $f(x)$ from $x = a$ to $x = b$



This thin strip is going to sweep the graph from $x = a$ to $x = b$ (remember $dx = \Delta x \rightarrow 0$)

- **The Fundamental Theorem of Calculus**

If $\frac{d}{dx}[g(x)] = f(x)$ This $\boxed{\int f(x)\,dx = g(x) + c}$

Then

$$\boxed{\int_a^b f(x)\,dx = [g(x)]_a^b = g(b) - g(a)}$$

Example 14

Evaluate the following integral $\int_1^2 (x^2 + x + 1)dx$

Solution

$$\int_1^2 (x^2 + x + 1)dx = \left[\frac{x^3}{3} + \frac{x^2}{2} + x\right]_1^2$$

$$= \left[\frac{(2)^3}{3} + \frac{(2)^2}{2} + (2)\right] - \left[\frac{(1)^3}{3} + \frac{(1)^2}{2} + (1)\right]$$

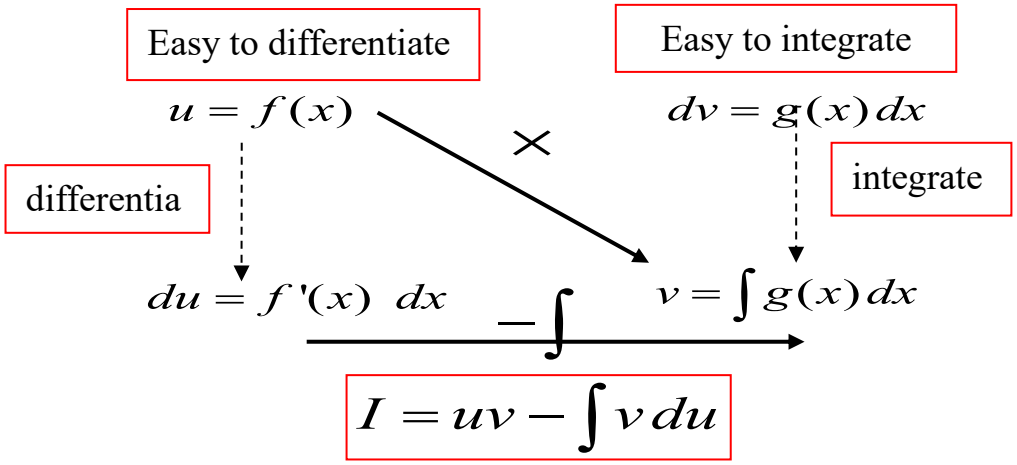
$$= 6.667 - 1.833$$

$$= 4.833$$

2.3. Integration by Parts

$$I = \int f(x)\, g(x)\; dx$$

We take



Example 1

$$I = \int x\cos x\,dx$$

$u = x$ $\quad dv = \cos x\,dx$

$du = dx$ $\quad v = \sin x$ (× diagonal, $-\int$ horizontal)

$$I = x\sin x - \int \sin x\,dx$$
$$= x\sin x - (-\cos x) + c$$
$$= x\sin x + \cos x + c$$

Example 2

$$I = \int x^2 e^{3x}\,dx$$

$u = x^2$ $\quad dv = e^{3x}\,dx$

$du = 2x\,dx$ $\quad v = \frac{1}{3}e^{3x}$ ($-\int$)

$$I = \frac{1}{3}x^2e^{3x} - \int 2x\frac{1}{3}e^{3x}\,dx$$
$$= \frac{1}{3}x^2e^{3x} - \frac{2}{3}\int xe^{3x}\,dx$$

$u = x$ $\quad dv = e^{3x}\,dx$

$du = dx$ $\quad v = \frac{1}{3}e^{3x}$ ($-\int$)

$$I = \frac{1}{3}x^2e^{3x} - \frac{2}{3}\left[\frac{1}{3}xe^{3x} - \int\frac{1}{3}e^{3x}\,dx\right]$$
$$= \frac{1}{3}x^2e^{3x} - \frac{2}{3}\left[\frac{1}{3}xe^{3x} - \frac{1}{9}e^{3x}\right] + c$$
$$= \frac{1}{3}x^2e^{3x} - \frac{2}{9}xe^{3x} + \frac{2}{27}e^{3x} + c$$

Example 3

$$I = \int x\,Ln\,x\,dx$$

$u = Ln\,x \qquad dv = x\,dx$

$\times$

$du = \frac{1}{x}dx \qquad v = \frac{x^2}{2}$

$-\int$

$$I = \frac{x^2}{2}Ln\,x - \int \frac{1}{x}\,\frac{x^2}{2}\,dx$$

$$= \frac{x^2}{2}Ln\,x - \int \frac{x}{2}\,dx$$

$$= \frac{x^2}{2}Ln\,x - \frac{x^2}{4} + c$$

Example 4

$$I = \int Ln\,x\,dx$$

$u = Ln\,x \qquad dv = dx$

$\times$

$du = \frac{1}{x}dx \qquad v = x$

$-\int$

$$I = x\,Ln\,x - \int \frac{1}{x}\,x\,dx$$

$$= x\,Ln\,x - \int dx$$

$$= x\,Ln\,x - x + c$$

Example 5

$$I = \int \sec^3 x\,dx$$

$$= \int \sec x\,\sec^2 x\,dx$$

$u = \sec x \qquad dv = \sec^2 x\,dx$

$\times$

$du = \sec x\tan x\,dx \qquad v = \tan x$

$-\int$

$$I = \sec x\tan x - \int \sec x\tan^2 x\,dx$$

$$= \sec x\tan x - \int \sec x(\sec^2 x - 1)\,dx$$

$$= \sec x\tan x - \int (\sec^3 x - \sec x)\,dx$$

$$I = \sec x \tan x - \left[I - Ln(\sec x + \tan x) \right]$$

$$I = \sec x \tan x - I + Ln(\sec x + \tan x)$$

Then

$$2I = \sec x \tan x + Ln(\sec x + \tan x) \quad \times \frac{1}{2}$$

$$\boxed{I = \frac{1}{2}\sec x \tan x + \frac{1}{2} Ln(\sec x + \tan x) + c}$$

Example 6

$$I = \int e^{2x} \sin 3x \, dx \qquad u = e^{2x} \quad dv = \sin 3x \, dx$$

$$\times \searrow$$

$$du = 2e^{2x} \, dx \qquad v = \frac{-1}{3}\cos 3x$$

$$\xrightarrow{-\int}$$

$$I = \frac{-1}{3} e^{2x} \cos 3x - \int \frac{-1}{3} 2e^{2x} \cos 3x \, dx$$

$$= \frac{-1}{3} e^{2x} \cos 3x + \frac{2}{3} \int e^{2x} \cos 3x \, dx$$

We have to do integration by parts again

$$u = e^{2x} \qquad dv = \cos 3x \, dx$$

$$\times \searrow$$

$$du = 2e^{2x} \, dx \qquad v = \frac{1}{3}\sin 3x$$

$$\xrightarrow{-\int}$$

$$I = \frac{-1}{3} e^{2x} \cos 3x + \frac{2}{3}\left[\frac{1}{3} e^{2x} \sin 3x - \int \frac{1}{3} 2e^{2x} \sin 3x \, dx \right]$$

$$= \frac{-1}{3} e^{2x} \cos 3x + \frac{2}{3}\left[\frac{1}{3} e^{2x} \sin 3x - \frac{2}{3} \int e^{2x} \sin 3x \, dx \right]$$

$$= \frac{-1}{3} e^{2x} \cos 3x + \frac{2}{3}\left[\frac{1}{3} e^{2x} \sin 3x - \frac{2}{3} I \right]$$

Then

$$I = \frac{-1}{3}e^{2x}\cos 3x + \frac{2}{9}e^{2x}\sin 3x - \frac{4}{9}I$$

$$I + \frac{4}{9}I = \frac{-1}{3}e^{2x}\cos 3x + \frac{2}{9}e^{2x}\sin 3x$$

$$\frac{13}{9}I = \frac{-1}{3}e^{2x}\cos 3x + \frac{2}{9}e^{2x}\sin 3x \times \frac{9}{13}$$

$$\boxed{I = \frac{-3}{13}e^{2x}\cos 3x + \frac{2}{13}e^{2x}\sin 3x + c}$$

Example

Calcula $I_n = \int \sin^n x\,dx$

Solution

$$I_n = \int \sin^n x\,dx = \int \sin^{n-1} x\ \sin x\,dx$$

$u = \sin^{n-1} x$ $\quad\times\quad$ $dv = \sin x\ dx$

$du = (n-1)\sin^{n-2} x\ \cos x\ dx$ $\quad -\int\quad$ $v = -\cos x$

$$I_n = -\cos x \sin^{n-1} x - \int -(n-1)\sin^{n-2} x\ \cos^2 x\,dx$$

$$I_n = -\cos x \sin^{n-1} x + (n-1)\int \sin^{n-2} x\,(1-\sin^2 x)\,dx$$

$$= -\cos x \sin^{n-1} x + (n-1)\int (\sin^{n-2} x - \sin^n x)\,dx$$

$$= -\cos x \sin^{n-1} x + (n-1)\left[I_{n-2} - I_n\right]$$

$$= -\cos x \sin^{n-1} x + (n-1)I_{n-2} - (n-1)I_n$$

Then

$$I_n + (n-1)I_n = -\cos x \sin^{n-1} x + (n-1)I_{n-2}$$

$$n I_n = -\cos x \sin^{n-1} x + (n-1)I_{n-2}$$

$$\boxed{I_n = \frac{-1}{n}\cos x\ \sin^{n-1} x + \frac{n-1}{n}I_{n-2}}$$

Important note

From the last relation we can calculate the

$$I_4 = \int \sin^4 x\,dx \qquad n = 4$$

$$I_n = \frac{-1}{n}\cos x \sin^{n-1} x + \frac{n-1}{n} I_{n-2}$$

Then $$I_4 = \frac{-1}{4}\cos x \sin^3 x + \frac{3}{4} I_2$$

Similarly, to calculate I_2

$$I_2 = \frac{-1}{2}\cos x \sin x + \frac{1}{2} I_0$$

And finally

$$I_0 = \int \sin^0 x\,dx = \int dx = x + c$$

Also from the last relation we can calculate the following integrals

$$I_3 = \int \sin^3 x\,dx \qquad n = 3 \qquad I_n = \frac{-1}{n}\cos x \sin^{n-1} x + \frac{n-1}{n} I_{n-2}$$

Then $$I_3 = \frac{-1}{3}\cos x \sin^2 x + \frac{2}{3} I_1$$

Where $$I_1 = \int \sin x\,dx$$

$$= -\cos x + c$$

2.4. Trigonometric Substitutions

For $\sqrt{a^2 - x^2}$ use $x = a\sin\theta$

Then we will get the following

$$\sqrt{a^2 - x^2} = \sqrt{a^2 - a^2\sin^2\theta} = a\sqrt{1 - \sin^2\theta} = a\sqrt{\cos^2\theta} = a\cos\theta$$

Similarl

For $\sqrt{a^2 + x^2}$ use $x = a\tan\theta$

$$\sqrt{a^2 + x^2} = \sqrt{a^2 + a^2\tan^2\theta} = a\sqrt{1 + \tan^2\theta} = a\sqrt{\sec^2\theta} = a\sec\theta$$

For $\sqrt{x^2 - a^2}$ use $x = a\sec\theta$

$$\sqrt{x^2 - a^2} = \sqrt{a^2\sec^2\theta - a^2} = a\sqrt{\sec^2\theta - 1} = a\sqrt{\tan^2\theta} = a\tan\theta$$

Example 8

$$\int \frac{1}{x^2\sqrt{4-x^2}}\,dx \qquad \text{Let} \quad x = 2\sin\theta \longrightarrow dx = 2\cos\theta\,d\theta$$

Then

$$\int \frac{1}{x^2\sqrt{4-x^2}}\,dx = \int \frac{1}{4\sin^2\theta\sqrt{4-4\sin^2\theta}}\,2\cos\theta\,d\theta$$

$$= \int \frac{1}{4\sin^2\theta\ 2\sqrt{1-\sin^2\theta}}\,2\cos\theta\,d\theta$$

$$= \int \frac{1}{4\sin^2\theta\ \sqrt{\cos^2\theta}}\cos\theta\,d\theta$$

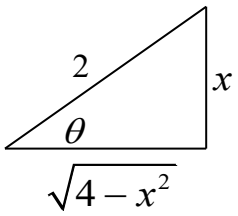
$$= \int \frac{1}{4\sin^2\theta\ \cos\theta}\cos\theta\,d\theta = \int \frac{1}{4\sin^2\theta}\,d\theta$$

Then $\int \frac{1}{x^2\sqrt{4-x^2}}\,dx = \frac{1}{4}\int \operatorname{cosec}^2\theta\,d\theta$

$$= -\frac{1}{4}\cot\theta + c$$

$$= -\frac{1}{4}\frac{\sqrt{4-x^2}}{x} + c$$

$x = 2\sin\theta \rightarrow \sin\theta = \frac{x}{2}$



$\cot\theta = \frac{\sqrt{4-x^2}}{x}$

Example 9

$$\int \frac{\sqrt{x^2-25}}{x}\,dx \qquad \text{Let} \quad x = 5\sec\theta \longrightarrow dx = 5\sec\theta\tan\theta\,d\theta$$

Then $\int \frac{\sqrt{x^2-25}}{x}\,dx = \int \frac{\sqrt{25\sec^2\theta-25}}{5\sec\theta}\,5\sec\theta\tan\theta\,d\theta$

$$= \int 5\sqrt{\sec^2\theta-1}\tan\theta\,d\theta$$

$$= \int 5\sqrt{\tan^2\theta}\tan\theta\,d\theta$$

$$= \int 5\tan\theta\tan\theta\,d\theta = \int 5\tan^2\theta\,d\theta$$

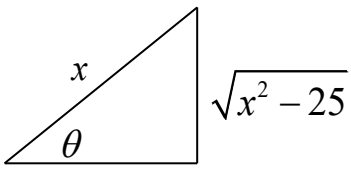
Then

$$\int \frac{\sqrt{x^2-25}}{x}\,dx = \int 5(\sec^2\theta - 1)\,d\theta$$

$$= 5(\tan\theta - \theta) + c$$

$$= 5\left[\frac{\sqrt{x^2-25}}{5} - \sec^{-1}\left(\frac{x}{5}\right)\right] + c$$

$$x = 5\sec\theta \rightarrow \cos\theta = \frac{5}{x}$$



$$\tan\theta = \frac{\sqrt{x^2-25}}{5}$$

2.5. Integration Using Partial Fractions

Example 1

Calculate $I = \int \frac{x-19}{x^2-3x-10}\,dx$

Solution

We have $\frac{x-19}{x^2-3x-10} = \frac{x-19}{(x+2)(x-5)} = \frac{A}{(x+2)} + \frac{B}{(x-5)}$

$$= \frac{A(x-5)+B(x+2)}{(x+2)(x-5)}$$

Then

$$x-19 = A(x-5) + B(x+2)$$

By $x = -2 \longrightarrow -21 = A(-7) + 0$ $\quad A = 3$

By $x = 5 \longrightarrow -14 = 0 + B(7)$ $\quad B = -2$

Then we have $\frac{x-19}{x^2-3x-10} = \frac{3}{(x+2)} + \frac{-2}{(x-5)}$

Hence $I = \int \frac{x-19}{x^2-3x-10}\,dx = \int \frac{3}{(x+2)} + \frac{-2}{(x-5)}\,dx$

$$= 3Ln(x+2) - 2Ln(x-5) + c$$

Example 2

Calculate $I=\int\frac{3x^2-7x-2}{x(x^2-1)}dx$

Solution

We have $\frac{3x^2-7x-2}{x(x^2-1)}=\frac{3x^2-7x-2}{x(x-1)(x+1)}=\frac{A}{x}+\frac{B}{(x-1)}+\frac{C}{(x+1)}$

Then $=\frac{A(x-1)(x+1)+Bx(x+1)+Cx(x-1)}{x(x-1)(x+1)}$

$3x^2-7x-2=A(x-1)(x+1)+Bx(x+1)+Cx(x-1)$

By $x=0 \longrightarrow -2=A(-1)(1)+0$ $\boxed{A=2}$

By $x=1 \longrightarrow -6=0+B(2)+0$ $\boxed{B=-3}$

By $x=-1 \longrightarrow 8=0+C(-1)(-2)$ $\boxed{C=4}$

Then we have $\frac{3x^2-7x-2}{x(x^2-1)}=\frac{2}{x}+\frac{-3}{(x-1)}+\frac{4}{(x+1)}$

Hence

$$I=\int\frac{3x^2-7x-2}{x(x^2-1)}dx=\int\frac{2}{x}+\frac{-3}{(x-1)}+\frac{4}{(x+1)}dx$$

$$=2Ln\,x-3Ln(x-1)+4Ln(x+1)+c$$

Example 3

Calculate $I=\int\frac{10}{(x-3)(x^2+1)}dx$

Solution

We have $\frac{10}{(x-3)(x^2+1)}=\frac{A}{(x-3)}+\frac{Bx+C}{(x^2+1)}=\frac{A(x^2+1)+(Bx+C)(x-3)}{(x-3)(x^2+1)}$

Then

$10=A(x^2+1)+(Bx+C)(x-3)$

By $x=3 \longrightarrow 10=A(10)+0$ $\boxed{A=1}$

By $x=0 \longrightarrow 10=(1)(1)+(C)(-3)$ $\boxed{C=-3}$

By $x=1 \longrightarrow 10=(1)(2)+(B-3)(-2)$ $\boxed{B=-1}$

Then we have $$\frac{10}{(x-3)(x^2+1)} = \frac{1}{(x-3)} + \frac{-x-3}{(x^2+1)}$$

Hence

$$I = \int \frac{10}{(x-3)(x^2+1)}dx = \int \frac{1}{(x-3)} + \frac{-x-3}{(x^2+1)}dx$$

$$= \int \frac{1}{(x-3)} - \frac{x}{(x^2+1)} - \frac{3}{(x^2+1)}dx$$

$$= Ln(x-3) - \frac{1}{2}Ln(x^2+1) - 3\tan^{-1}x + c$$

Example 4

Calculate $$I = \int \frac{5x^2+20x+6}{x(x+1)^2}dx$$

Solution

We have $$\frac{5x^2+20x+6}{x(x+1)^2} = \frac{A}{x} + \frac{B}{(x+1)} + \frac{C}{(x+1)^2}$$

Then $$= \frac{A(x+1)^2 + Bx(x+1) + Cx}{x(x+1)^2}$$

$$5x^2+20x+6 = A(x+1)^2 + Bx(x+1) + Cx$$

By $x=0 \longrightarrow 6 = A(1) + 0$ $\quad A=6$

By $x=-1 \longrightarrow -9 = 0 + C(-1)$ $\quad C=9$

By $x=1 \longrightarrow 31 = (6)(4) + B(1)(2) + (9)(1)$ $\quad B=-1$

Then we have $$\frac{5x^2+20x+6}{x(x+1)^2} = \frac{6}{x} + \frac{-1}{(x+1)} + \frac{9}{(x+1)^2}$$

Hence

$$I = \int \frac{5x^2+20x+6}{x(x+1)^2}dx = \int \frac{6}{x} - \frac{1}{(x+1)} + \frac{9}{(x+1)^2}dx$$

$$= \int \frac{6}{x} - \frac{1}{(x+1)} + 9(x+1)^{-2}dx$$

$$= 6Ln\,x - Ln(x+1) + 9\frac{(x+1)^{-1}}{-1} + c$$

Sheet [2]

Problem 1

Calculate the following integrals

(1) $\int x^6 - \sqrt{x} + \frac{1}{x^4} + 8 \, dx$

(2) $\int x^{3/2} + \frac{1}{x} + 9 \, dx$

(3) $\int x^2 \left(2x^3 - 5\right)^4 \, dx$

(4) $\int \left(5x + 3\right)^{10} \, dx$

(5) $\int \left(3x^2 - 7\right)^2 \, dx$

(6) $\int \left(3x - \frac{1}{x}\right)^2 \, dx$

(7) $\int \sin^9 x \cos x \, dx$

(8) $\int \cos^4 x \sin x \, dx$

(9) $\int \frac{x}{5x^2 - 2} \, dx$

(10) $\int \frac{x^2}{7x^3 + 9} \, dx$

(11) $\int \operatorname{cosec} x \, dx$

(12) $\int e^{\sinh x} \cosh x \, dx$

(13) $\int x e^{3x^2-1} \, dx$

(14) $\int e^{Lnx} \, dx$

(15) $\int x \cos(5x^2 - 7) \, dx$

(16) $\int \sin(3x + 8) \, dx$

(17) $\int \tan^2 x \, dx$

(18) $\int x\sqrt{x+2} \, dx$

(19) $\int (x+2)\sqrt{x+1} \, dx$

(20) $\int \frac{e^x}{1-e^x} \, dx$

(21) $\int \frac{e^{-2x}}{1+e^{-2x}} \, dx$

(22) $\int \sec 2x \, dx$

(23) $\int \frac{1}{16+x^2}\,dx$

(24) $\int \frac{1}{9+(x+5)^2}\,dx$

(25) $\int \frac{1}{\sqrt{9-x^2}}\,dx$

(26) $\int \frac{1}{\sqrt{(x-4)^2+25}}\,dx$

Problem 2

Calculate the following integrals

(1) $\int x \sin 2x\,dx$

(2) $\int x^2 \cos x\,dx$

(3) $\int x^3 e^{2x}\,dx$

(4) $\int x^2 Ln\,x\,dx$

(5) $\int Ln\,x\,dx$

(6) $\int \tan^{-1} x\,dx$

(7) $\int x\sqrt{x+1}\,dx$

(8) $\int x\tan^{-1}x\,dx$

(9) $\int \frac{Ln\,x}{x}\,dx$

(10) $\int x^n Ln\,x\,dx \quad n\neq -1$

(11) $\int e^{2x}\cos 3x\,dx$

(12) $\int \operatorname{cosec}^3 x\,dx$

(13) If $I_n=\int x^n e^x\,dx$ prove that,

$$I_n = x^n e^x + n\,I_{n-1}$$

Then use this result to calculate

(i) $I_4=\int x^4 e^x\,dx$ (ii) $I_5=\int x^5 e^x\,dx$

(14) If $I_n=\int \cos^n x\,dx$ prove that,

$$I_n=\frac{1}{n}\sin x\,\cos^{n-1}x+\frac{n-1}{n}I_{n-2}$$

Then use this result to calculate

(i) $I_4 = \int \cos^4 x\,dx$ (ii) $I_5 = \int \cos^5 x\,dx$

(15) $\int \frac{1}{x^2\sqrt{9-x^2}}\,dx$

(16) $\int \frac{x^2}{\sqrt{16-x^2}}\,dx$

(17) $\int \sqrt{4-x^2}\,dx$

(18) $\int \frac{x^2}{\sqrt{9+x^2}}\,dx$

(19) $\int \frac{\sqrt{x^2-36}}{x}\,dx$

(20) $\int \frac{\sqrt{x^2-4}}{x^2}\,dx$

(21) $\int \frac{1}{x^2+x-2}\,dx$

(22) $\int \frac{x-5}{x^2-1}\,dx$

(23) $\int \frac{5x-2}{x^2-4}\,dx$

(24) $\int \frac{6x}{x^2-x-2}\,dx$

(25) $\int \frac{3x}{x^2-3x-4}\,dx$

(26) $\int \frac{2x^2-5x+2}{x^3+x}\,dx$

(27) $\int \frac{4x-5}{x^3-3x^2}\,dx$

(28) $\int \frac{x+2}{x^3+x}\,dx$

(29) $\int \frac{x-1}{(x-4)(x-3)^2}\,dx$

(30) $\int \frac{2x+3}{x^2+2x+1}\,dx$

(31) $\int \frac{x^3+x}{x^2-1}\,dx$

(32) $\int \frac{2x^3-4x^2-15x+5}{x^2-2x-8}\,dx$

Chapter (3) CIRCLE AND CONIC SECTIONS

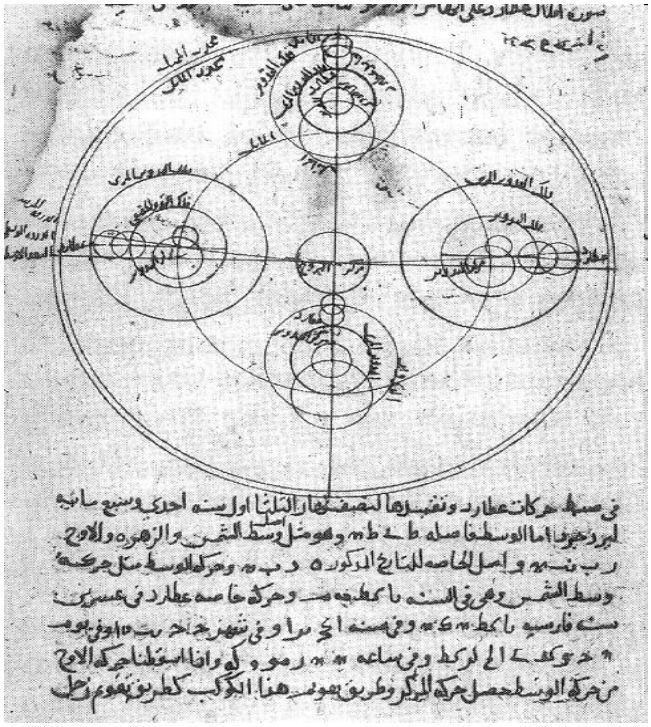
3.1. THE CIRCLE

- **Historical Notes About**

The word "circle" derives from the Greek, *kirkos* "a circle," from the base *ker-* which means to turn or bend. The origins of the words “circus” and “circuit” are closely related.
The circle has been known since before the beginning of recorded history. Natural circles would have been observed, such as the Moon, Sun, and a short plant stalk blowing in the wind on sand, which forms a circle shape in the sand. The circle is the basis for the wheel, which, with related inventions such as gears, makes much of modern civilization possible. In mathematics, the study of the circle has helped in inspiring the development of geometry, astronomy, and calculus.

Some highlights in the history of the circle are:

- 1700 BC – The Rhind papyrus gives a method to find the area of a circular field.
- 300 BC – Book 3 of Euclid’s Elements deals with the properties of circles.
- 1880 – Lindemann proves that π is transcendental, effectively settling the millennia-old problem of squaring the circle



Circles on an old Arabic astronomical drawing

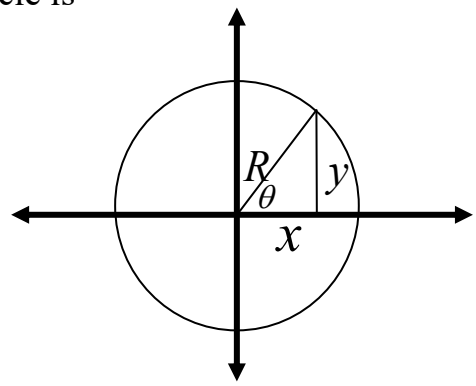
- **The Equation of the Circle**

The equation of the circle **with center at the origin** and radius R is

$$x^2 + y^2 = R^2$$

The parametric equation of that circle is

$$x = R\cos\theta$$
$$y = R\sin\theta$$

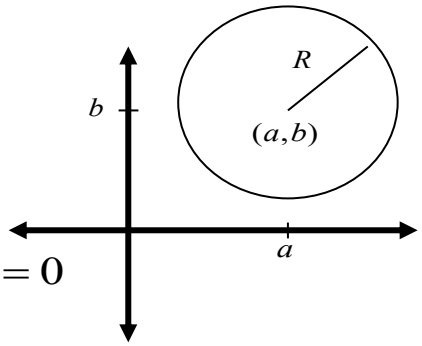


The equation of the circle **with center at (a,b)** and radius R is

$$(x-a)^2 + (y-b)^2 = R^2$$

Then

$$x^2 - 2ax + a^2 + y^2 - 2by + b^2 = R^2$$
$$x^2 + y^2 - 2ax - 2by + a^2 + b^2 - R^2 = 0$$



Or

$$x^2 + y^2 + 2hx + 2ky + d = 0 \longrightarrow \begin{matrix} h = -a \quad k = -b \\ d = a^2 + b^2 - R^2 \end{matrix}$$

Then for the circle

$$x^2 + y^2 + 2hx + 2ky + d = 0$$

The center is $(-h, -k)$

Also $$R = \sqrt{a^2 + b^2 - d} = \sqrt{(-h)^2 + (-k)^2 - d}$$

The radius is $$R = \sqrt{h^2 + k^2 - d}$$

Example 1

Find the equation of the circle whose center is at (2,-3) and with radius 4

Solution

$$(x-2)^2+(y+3)^2=16$$

Then

$$x^2-4x+4+y^2+6y+9=16$$
$$x^2+y^2-4x+6y-3=0$$

Important Note

For the circle $x^2+y^2-4x+6y-3=0$ we have

$$2h=-4 \quad 2k=6 \quad d=-3$$

Then $h=-2 \quad k=3 \quad d=-3$

The center $(2,-3)$

The radius $R=\sqrt{4+9+3}=\sqrt{16}=4$

Example 2

Study the circle $x^2+y^2+8x-10y+5=0$

Solution

$$2h=8 \quad 2k=-10 \quad d=5$$

Then $h=4 \quad k=-5 \quad d=5$

The center $(-4,5)$

The radius $R=\sqrt{16+25-5}=\sqrt{36}=6$

- **Completing The Squares**

Another solution by using the method of ***Completing The Squares***

For the circle

$$x^2 + y^2 + 8x - 10y + 5 = 0$$

Step 1: gather the similar terms

$$(x^2 + 8x) + (y^2 - 10y) + 5 = 0$$

Step 2: take the square root of the squared terms

$$(x \quad)^2 + (y \quad)^2 \qquad + 5 = 0$$

Step 3: take half the coefficients of the linear terms and then put both brackets
to be complete squares (we still have another step)

$$(x+4)^2 + (y-5)^2 \qquad + 5 = 0$$

Step 4: square the numbers inside the brackets and then subtract them as follows

$$(x+4)^2 + (y-5)^2 - 16 - 25 + 5 = 0$$

Thus

$$(x+4)^2 + (y-5)^2 = 36 \longrightarrow \begin{matrix} (a,b) = (-4,5) \\ R = 6 \end{matrix}$$

Example 3

Study the circle $x^2 + y^2 - 6x + 3y + 1 = 0$

Solution

Buy completing the square we get

$$(x^2 - 6x) + (y^2 + 3y) + 1 = 0$$

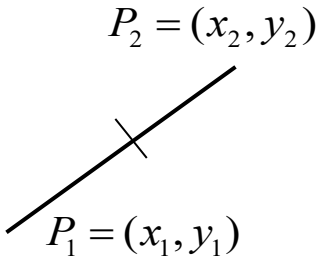
$$(x-3)^2 + (y + \frac{3}{2})^2 - 9 - \frac{9}{4} + 1 = 0$$

$$(x-3)^2 + (y + \frac{3}{2})^2 = \frac{41}{4} \longrightarrow \boxed{(a,b) = (3, \frac{-3}{2}),\ R = \frac{\sqrt{41}}{2}}$$

Note

For the points $P_1 = (x_1, y_1)$, $P_2 = (x_2, y_2)$

The mid-point will be $= \dfrac{(x_1, y_1) + (x_2, y_2)}{2}$



The distance between the points will $= \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$

Example 4

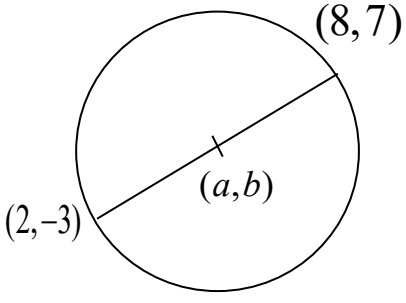
Find the equation of the circle on which the points (2,-3) and (8,7) are the end points of a diameter of this circle

Solution

The center will be $(a,b) = \dfrac{(2,-3) + (8,7)}{2} = (5,2)$

The radius will be

$$R = \frac{1}{2}\sqrt{(2-8)^2 + (-3-7)^2} = \frac{\sqrt{136}}{2}$$



The equation of the circle will be

$$\boxed{(x-5)^2 + (y-2)^2 = \frac{136}{4}}$$

3.2. The Parabola

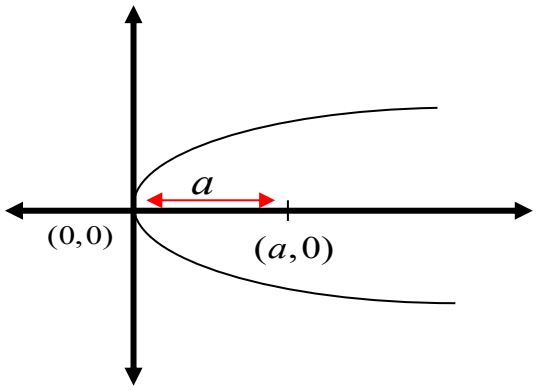
The equation of the parabola with vertex at the origin and focus on the positive X-axis is

$$y^2 = 4ax$$

The vertex will be at $(0,0)$

The distance between the

The focus will be at $(a,0)$



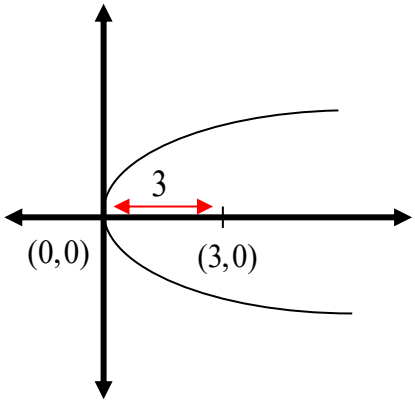
Example 1

$$y^2 = 12x$$

then $4a = 12 \longrightarrow a = 3$

The vertex will be at $(0,0)$

The focus will be at $(3,0)$

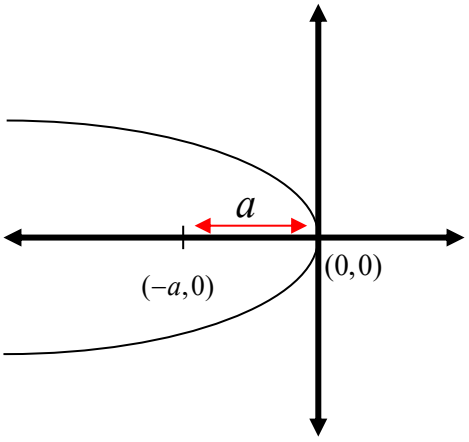


The equation of the parabola with vertex at the origin and focus on the negative X-axis is

$$y^2 = -4ax$$

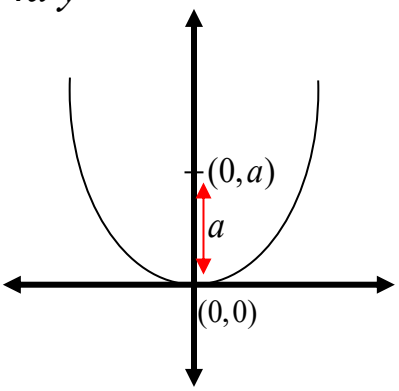
The vertex will be $(0,0)$

The focus will be $(-a,0)$



Also we have the following cases of the parabola with vertex at the origin and focus on the Y-axis *(Please compare the four cases)*

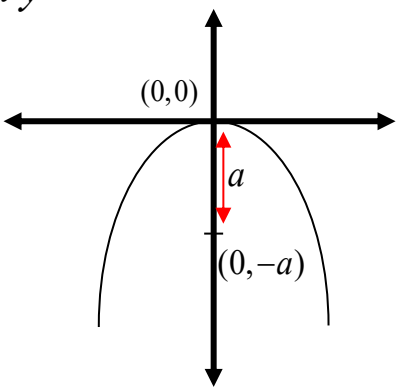
$x^2 = 4a\,y$



The vertex will be at $(0,0)$

The focus will be at $(0,a)$

$x^2 = -4a\,y$



The vertex will be at $(0,0)$

The focus will be at $(0,-a)$

Example 2

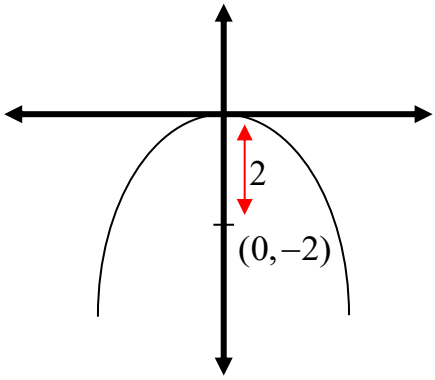
$$x^2 = -8\,y$$

then

$$4a = 8 \longrightarrow \boxed{a = 2}$$

The vertex will be at $(0,0)$

The focus will be at $(0,-2)$



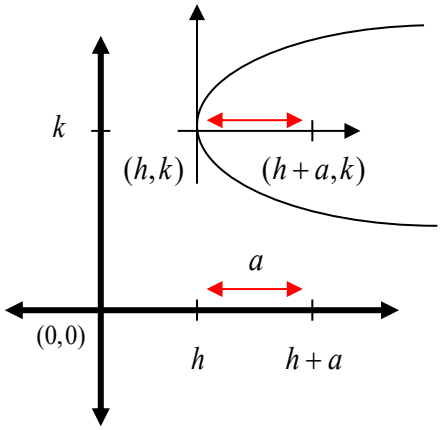
- **Some General Cases of Parabolas**

For the equation

$$(y-k)^2 = 4a(x-h)$$

The vertex will be at (h,k)

The focus will be at $(h+a,k)$



Example 3

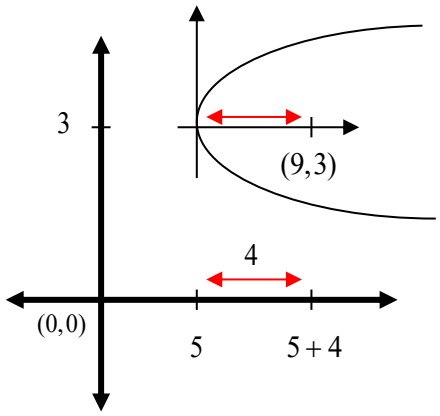
$$(y-3)^2 = 16(x-5)$$

then $4a = 16 \rightarrow \boxed{a = 4}$

also $\boxed{h = 5 \ , \ k = 3}$

The vertex will be at $(5,3)$

The focus will be at $(5+4,3) = (9,3)$



Example 4

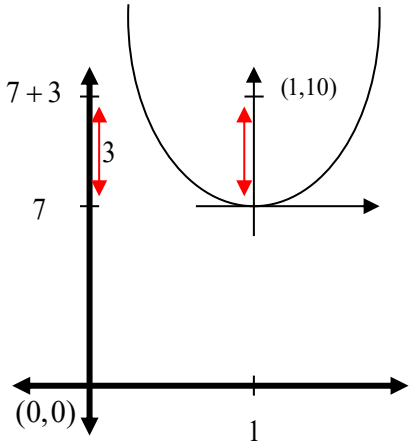
$$(x-1)^2 = 12(y-7)$$

then $4a = 12 \rightarrow \boxed{a = 3}$

als $\boxed{h = 1 \ , \ k = 7}$

The vertex will be at $(1,7)$

The focus will be at $(1,7+3) = (1,10)$



Example 5

Study the equation of the $x^2 - 2x - 12y + 85 = 0$

Solution

By using the method of completing of the square we get

$$x^2 - 2x - 12y + 85 = 0$$

$$(x^2 - 2x) - 12y + 85 = 0$$

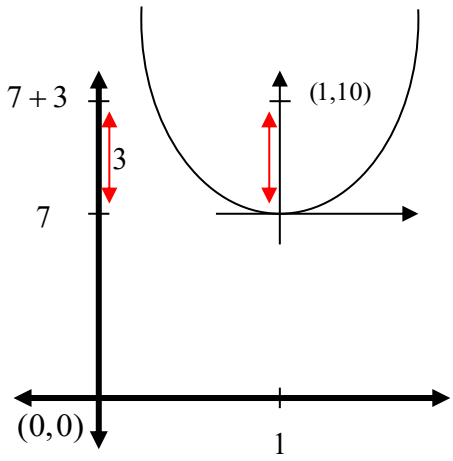
$$(x-1)^2 - 1 - 12y + 85 = 0$$

$$(x-1)^2 = 12y - 84$$

$$(x-1)^2 = 12(y-7)$$

We will study the $(x-1)^2 = 12(y-7)$

$$(x-1)^2 = 12(y-7)$$



then $4a = 12 \longrightarrow \boxed{a = 3}$

also $\boxed{h = 1 \ , \ k = 7}$

The vertex will be at $(1,7)$

The focus will be at $(1,7+3) = (1,10)$

Example 6

Study the equation of $y^2 + 4y + 4x - 16 = 0$

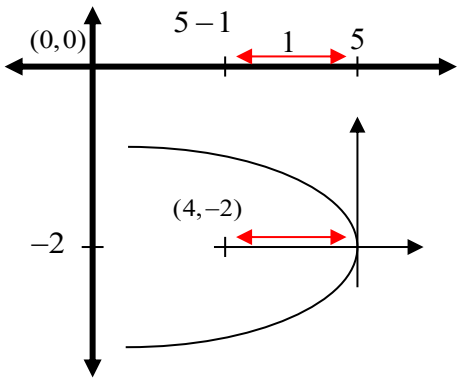
Solution

By using the method of completing of the square we get

$$(y^2 + 4y) + 4x - 16 = 0$$
$$(y+2)^2 - 4 + 4x - 16 = 0$$
$$(y+2)^2 = -4x + 20$$
$$(y+2)^2 = -4(x-5)$$

We will study the equation $(y+2)^2 = -4(x-5)$

$$(y+2)^2 = -4(x-5)$$



then $4a = 4 \longrightarrow \boxed{a = 1}$

also $\boxed{h = 5 \ , \ k = -2}$

The vertex will be at $(5,-2)$

The focus will be at $(5-1,-2) = (4,-2)$

Example 7

Find the equation of the parabola whose vertex is at (-2,5) and its focus Is at the point (-2,8)

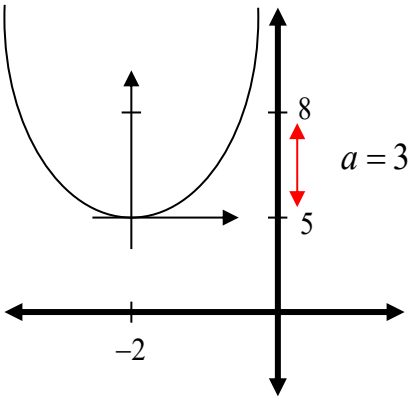
Solution

By sketching the parabola first

Then we can assume the equation to be on the form

$$(x-h)^2 = 4a(y-k)$$

but $h=-2\ ,\ k=5\ ,\ a=3$



Then the equation is

$$(x+2)^2 = 12(y-5)$$

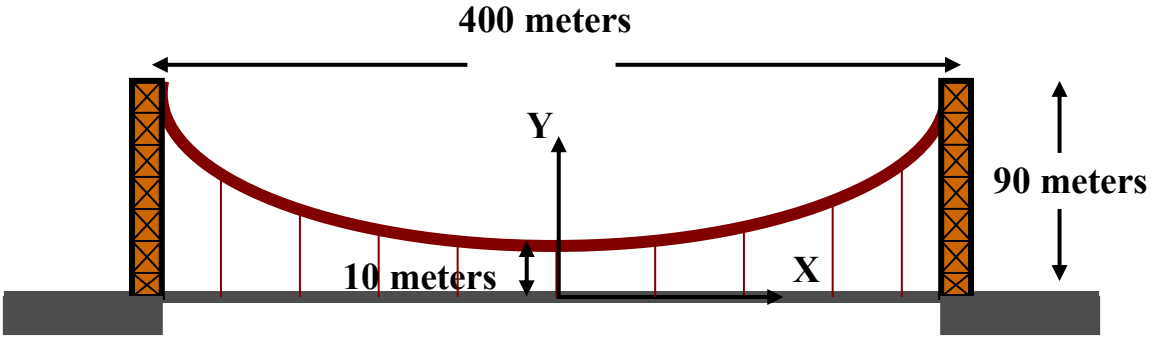
Example 8

One section of a suspended bridge has its weight uniformly distributed between twin towers that are 400 meters apart and that rise 90 meters above the horizontal roadway.

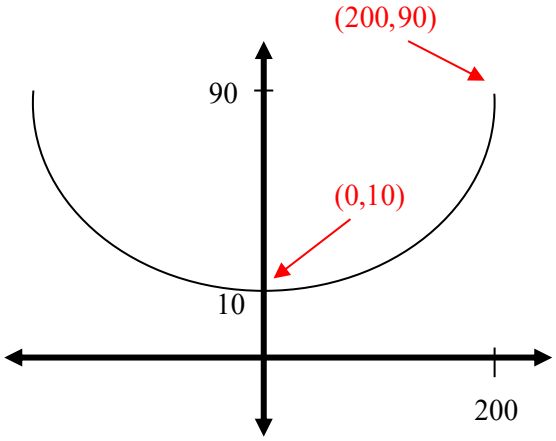
A cable strung between the tops of the towers has the shape of a parabola, with center point 10 meters above the horizontal roadway. Suppose coordinate axes are introduced as shown in the figure.

(a) Find the equation of the parabola

(b) Find the height of the cable at a distance 100 meters from its center



Solution



We can assume

$(x-h)^2 = 4a(y-k)$

then $\boxed{h=0\ ,\ k=10}$

The equation will take the form

$x^2 = 4a(y-10)$

The point $(x,y)=(200,90)$ satisfies the

$(200)^2 = 4a(90-10)$

Then

$40000 = 4a(80)$

$500 = 4a$

The equation will be

$\boxed{x^2 = 500(y-10)}$

Now at a distance 100 meters from the center we have $x=100$

$\boxed{x=100} \longrightarrow (100)^2 = 500(y-10)$

$10000 = 500(y-10)$

$20 = (y-10) \longrightarrow \boxed{y=30}$

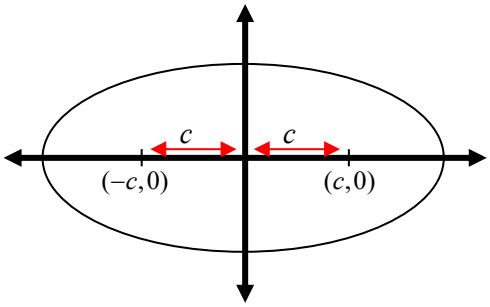
Then at a distance 100 meters from the center of the bridge the height of the cable is 30 meters

3.3. THE ELLIPSE

The equation of the Ellipse with center at the origin and foci on the X-axis is

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1 \quad , a^2>b^2$$

The center will be at $(0,0)$



The distance between the

$$c^2=a^2-b^2$$

The foci will be at $(c,0)\ ,(-c,0)$

Example 1

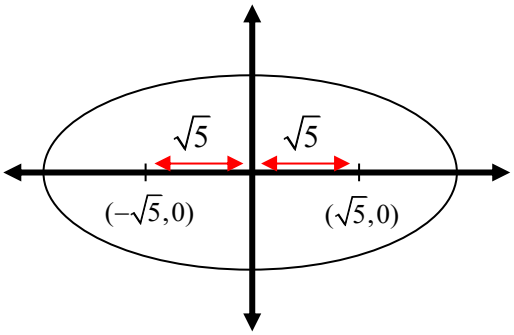
$$\frac{x^2}{9}+\frac{y^2}{4}=1$$

The center $(0,0)$

Then $a^2=9>b^2=4$

$$\longrightarrow \quad c^2=9-4=5$$

Then $c=\sqrt{5}$



The foci will be at $(\sqrt{5},0)\ ,(-\sqrt{5},0)$

Remark 1

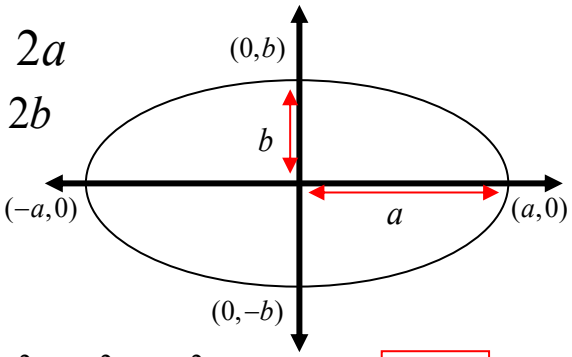
For the Ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1\ , a^2>b^2$

The length of the major axis is $2a$

The length of the minor axis is $2b$

The eccentricity is

$$e=\sqrt{1-\frac{b^2}{a^2}}<1$$

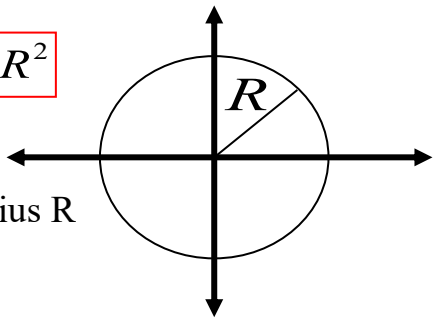


Noting that $e^2=1-\frac{b^2}{a^2}=\frac{a^2-b^2}{a^2}=\frac{c^2}{a^2}$ $\longrightarrow$ $e=\frac{c}{a}$

Remark 2

In the case of $a = b = R$ we get

$$\frac{x^2}{R^2} + \frac{y^2}{R^2} = 1 \longrightarrow \boxed{x^2 + y^2 = R^2}$$



This is a circle with center at the origin and radius R

In this case

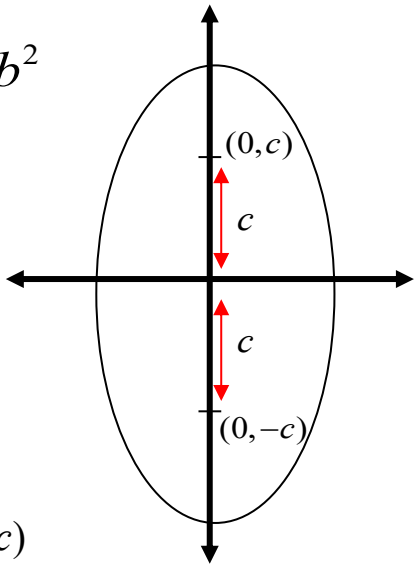
$$e = \sqrt{1 - \frac{b^2}{a^2}} = \sqrt{1 - \frac{R^2}{R^2}} = 0$$

- **Another Special Case of the Ellipse**

  The equation of the Ellipse with center at the origin and foci on the Y-axis is

$$\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1 \quad , a^2 > b^2$$

The center will be at $(0,0)$



The distance between the center and each focus is c where

$$c^2 = a^2 - b^2$$

The foci will be at $(0,c)\ ,\ (0,-c)$

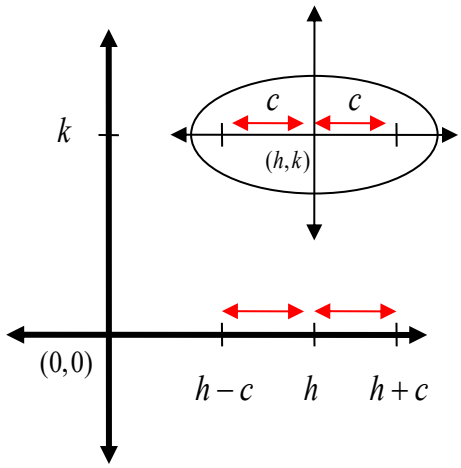
- **More Cases of The Ellipse**

$$\frac{(x-h)^2}{a^2}+\frac{(y-k)^2}{b^2}=1\ ,\ a^2>b^2$$

The center will be at (h,k)

The foci will be at $(h+c,k)\ ,\ (h-c,k)$

where $c^2=a^2-b^2$



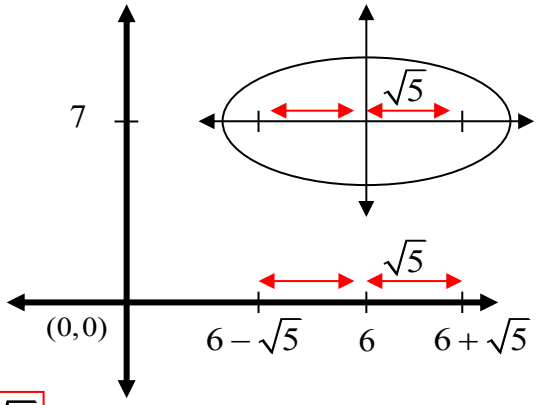
Example 2

$$\frac{(x-6)^2}{9}+\frac{(y-7)^2}{4}=1$$

The center will be at $(6,7)$

Then $a^2=9>b^2=4$

$c^2=9-4=5$ then $c=\sqrt{5}$



The foci will be at $(6+\sqrt{5},7)\ ,\ (6-\sqrt{5},7)$

Example 3

Study the equation of the Ellipse $9x^2+16y^2-18x+64y-71=0$

solution

By completing the square we get

$$(9x^2-18x)+(16y^2+64y)-71=0$$

$$9(x^2-2x)+16(y^2+4y)-71=0$$

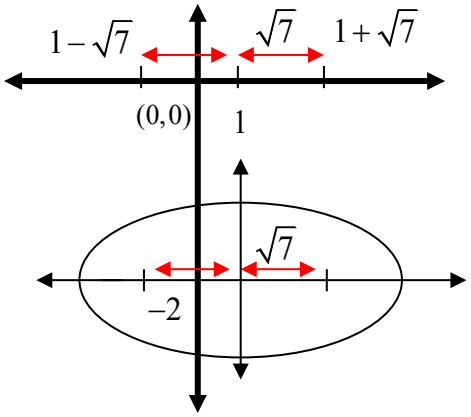
$$9\left[(x-1)^2-1\right]+16\left[(y+2)^2-4\right]-71=0$$

$$9(x-1)^2-9+16(y+2)^2-64-71=0$$

$$9(x-1)^2+16(y+2)^2=144 \longleftarrow \div 144$$

$$\frac{9(x-1)^2}{144}+\frac{16(y+2)^2}{144}=1$$

$$\frac{(x-1)^2}{16}+\frac{(y+2)^2}{9}=1$$



The center will be at $(1,-2)$

Then $a^2=16>b^2=9$

$c^2=16-9=7$ then $c=\sqrt{7}$

The foci will be at $(1+\sqrt{7},-2)\,,(1-\sqrt{7},-2)$

Example 4

Study the equation of the Ellipse

$$25x^2+4y^2-250x-16y+541=0$$

solution

By completing the square we get

$$(25x^2-250x)+(4y^2-16y)+541=0$$

$$25(x^2-10x)+4(y^2-4y)+541=0$$

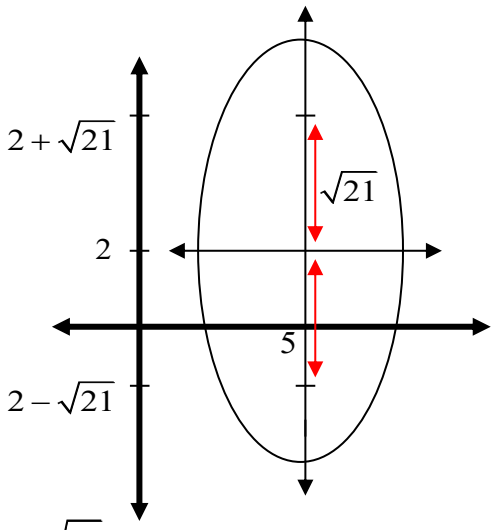
$$25\left[(x-5)^2-25\right]+4\left[(y-2)^2-4\right]+541=0$$

$$25(x-5)^2-625+4(y-2)^2-16+541=0$$

$$25(x-5)^2+4(y-2)^2=100 \quad \leftarrow \div 100$$

$$\frac{25(x-5)^2}{100}+\frac{4(y-2)^2}{100}=1$$

$$\frac{(x-5)^2}{4}+\frac{(y-2)^2}{25}=1$$



The center will be at $(5,2)$

Then $a^2=25>b^2=4$

$c^2=25-4=21$ then $c=\sqrt{21}$

The foci will be at $(5,2+\sqrt{21})\,,(5,2-\sqrt{21})$

Example 5

Find the equation of the Ellipse with vertices $(\pm 8,0)$ and foci $(\pm 5,0)$

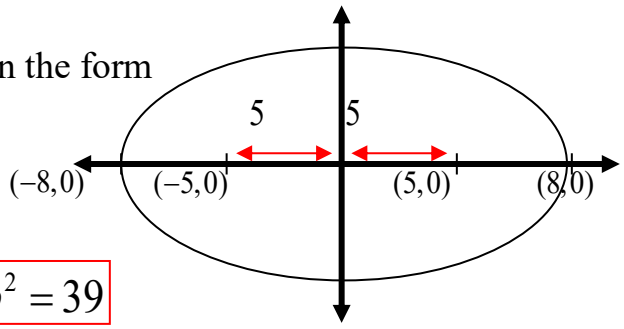
Solution

By sketching the given data about the Ellipse we get

$\boxed{a=8}$ and $\boxed{c=5}$

We can assume the equation on the form

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$$



Also $\boxed{\begin{aligned} c^2 &= a^2-b^2 \\ 25 &= 64-b^2 \end{aligned}} \longrightarrow \boxed{b^2=39}$

Then the equation will be $\dfrac{x^2}{64}+\dfrac{y^2}{39}=1$

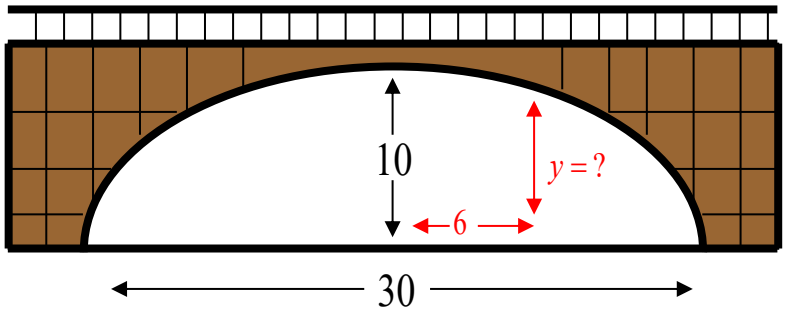
Example 6

The arch of a bridge is semi-elliptical, with the major axis horizontal. The base of the arch is 30 meters across and the highest part is 10 meters above the horizontal roadway. Find the height of the arch 6 meters from the center of the base.

Solution

We have

$\boxed{\begin{aligned} a &= 15 \\ b &= 10 \end{aligned}}$



The equation of this Ellipse will be $\dfrac{x^2}{225}+\dfrac{y^2}{100}=1$

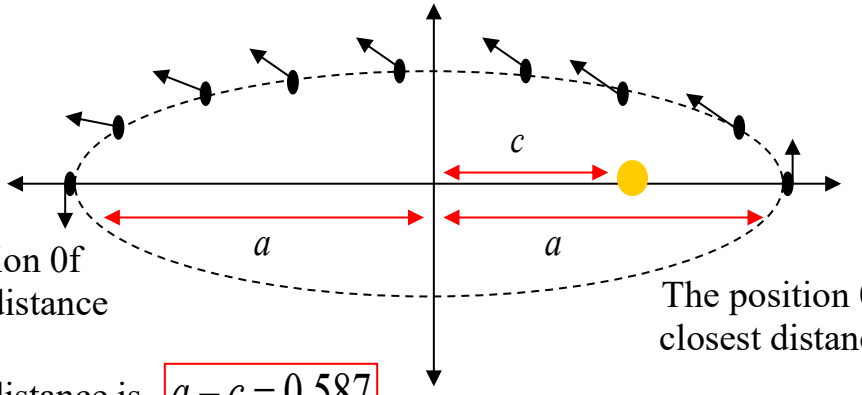
At $x=6$ we have $\dfrac{36}{225}+\dfrac{y^2}{100}=1 \longrightarrow \boxed{y=\sqrt{84}\approx 9.165}$

Example 7

Halley's comet has an elliptical orbit with the sun at one of its foci its eccentricity is e=0.967. The closest distance between the comet and the sun is 0.587 AU (Astronomical Unit). Approximate the maximum distance of the comet from the sun.

Note: 1 AU = 93,000,000 miles = 150,000,000 kilometers

Solution



The closest distance is $a-c=0.587$

The maximum distance is $a+c=?$

Also $e=\frac{c}{a}$

Then $c=ae$

By using this result with $a-c=0.587$

$$a-ae=0.587$$

$$a(1-e)=0.587$$

$$a=\frac{0.587}{1-e}$$

But $e=0.967$ $\longrightarrow$ $a=17.788$, $c=17.2$

Then the maximum distance is

$$a+c=17.788+17.2=34.988\approx 35\ \ AU$$

Sheet [3]

(1) Find the equation of the line passing through the point $(5,-3)$ and its slop is -4.

(2) Find the equation of the line passing through the points $(1,-1)$, $(-4,7)$

(3) Find the equation of the line passing through the point $(-4,3)$ and parallel to the line $5x-2y-4=0$.

(4) Find the equation of the line passing through the point $(3,2)$ and perpendicular to the line $4x+5y+1=0$.

(5) Find the equation of the line passing through the point $(-3,8)$ and through the point of intersection of the lines $x-y+1=0$, $2x+y=4$ and then sketch the graph of the line.

(6) Find the equation of the circle with center $(3,2)$ and radius 4.

(7) Find the equation of the circle in which the points $(1,3)$, $(5,-7)$ are the end points of one of its diameters.

(8) Find the center and radius of the circle $x^2+y^2-8x+6y-2=0$.

(9) By two different methods, find the center and radius of the circle $x^2+y^2+4x-8y-5=0$.

(10) Find the equation of the circle passing through the origin and through the points $(4,0)$, $(0,8)$

(11) Study and sketch the following Parabolas:

(a) $y^2 = -24x$

(b) $x^2 = 3y$

(c) $(x-4)^2 = 8(y+7)$

(d) $(x+1)^2 = -12(y-2)$

(e) $(y-1)^2 = 16(x+5)$

(f) $(y+3)^2 = -6(x-7)$

(g) $x^2 - 4x - y + 2 = 0$

(h) $x^2 + 4x + 4y - 16 = 0$

(i) $y^2 - 2y - 12x + 85 = 0$

(j) $y^2 - 20y - 6x + 100 = 0$

(12) Find the equation of the Parabola with vertex at the point $(-1,0)$ and focus at $(-4,0)$.

(13) Find the equation of the Parabola with vertex at the point $(1,-2)$ and focus at $(1,0)$.

(14) Find the equation of the Parabola with vertex at the point $(-3,5)$ with its axis parallel to the X-axis and passing through the point $(5,9)$.

(15) A parabolic flashlight reflector has the shape $x = 12y^2$. Where should the lightbulb be placed

(16) A parabolic satellite dish has the shape $y = 28x^2$. Where should the microphone be placed

(17) Study and sketch the following Ellipses:

(a) $\frac{x^2}{25}+\frac{y^2}{16}=1$

(b) $9x^2+4y^2=36$

(c) $\frac{(x-4)^2}{9}+\frac{(y+7)^2}{16}=1$

(d) $\frac{(x-3)^2}{8}+\frac{(y-2)^2}{36}=1$

(e) $4x^2+9y^2-32x-36y+64=0$

(f) $x^2+2y^2+2x-20y+43=0$

(g) $25x^2+4y^2-250x-16y+541=0$

(h) $9x^2+4y^2+18x+24y+9=0$

(18) Find the equation of the Ellipse with vertices at the points $(\pm 8,0)$ and foci at $(\pm 5,0)$.

(19) Find the equation of the Ellipse with foci at the points $(\pm 3,0)$ and its minor axis is of length 2.

Chapter (4) INFINITE SERIES

4.1 SERIES OF REAL NUMBERS

- **Geometric Series**

The geometric series is on the

$$\sum_{n=1}^{\infty} a r^{n-1} = a + ar + ar^2 + ar^3 + \cdots$$

a is the first

r is the base

If $-1 < r < 1$ then the sum of this series

$$\sum_{n=1}^{\infty} a r^{n-1} = \frac{a}{1-r}$$

Example

Find the sum of the following series $\sum_{n=1}^{\infty} 5(\frac{1}{3})^{n-1}$

Solution

This is a geometric series in which $a = 5$ $r = \frac{1}{3} < 1$

Then

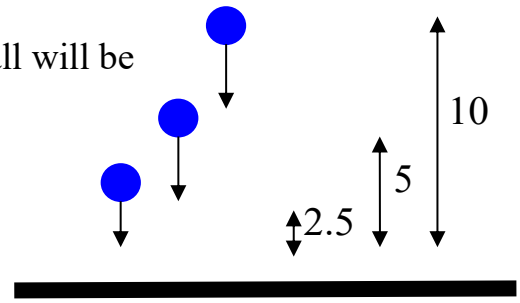
$$\sum_{n=1}^{\infty} 5(\frac{1}{3})^{n-1} = \frac{a}{1-r} = \frac{5}{1-1/3} = \frac{15}{2}$$

Example

A rubber ball is dropped from a height of 10 meters. If it rebounds one-half the distance after each fall, use the geometric series to approximate the total distance the ball travels before coming to rest.

Solution

The total distance traveled with the ball will be



$$10+10(\frac{1}{2})+10(\frac{1}{2})+10(\frac{1}{2})^2+10(\frac{1}{2})^2+\cdots$$

This is as the ball goes up and down after the first fall

The total distance is

$$10+\left[10(\frac{1}{2})+10(\frac{1}{2})+10(\frac{1}{2})^2+10(\frac{1}{2})^2+\cdots\right]$$

$$=10+\left[2\times10(\frac{1}{2})+2\times10(\frac{1}{2})^2+\cdots\right]$$

$$=10+\left[10+10(\frac{1}{2})+10(\frac{1}{2})^2+\cdots\right] \rightarrow a=10 \quad r=\frac{1}{2}<1$$

$$=10+\frac{10}{1-1/2}$$

$$=10+20=30$$

4.2 TESTS OF CONVERGENCE OF SERIES

- **Convergence and Divergence of Series**

Important Remake:

For the series $\sum_{n=1}^{\infty} x_n$ we have

If $\lim_{n\to\infty} x_n \neq 0$ then the series is divergent

If $\lim_{n\to\infty} x_n = 0$ then we have to do a test to decide whether the series is convergent or divergent

Example 1

$$\sum_{n=1}^{\infty} \frac{4n^2 - 2n + 3}{3n^2 - 5n - 1}$$ is divergent because

$$\lim_{n\to\infty} \frac{4n^2 - 2n + 3}{3n^2 - 5n - 1} = \frac{4}{3} \neq 0$$

Example 2

$$\sum_{n=1}^{\infty} \left(1 + \frac{1}{n}\right)^{5n}$$ is divergent because

$$\lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^{5n} = e^5 \neq 0$$

Important Note

A very important type of series whose behavior is well known is the p-series

$$\sum_{n=1}^{\infty} \frac{1}{n^p}$$

If $p \le 1$ the series is divergent

If $p > 1$ the series is convergent

Example 3

$\sum_{n=1}^{\infty} \frac{1}{n^{2/3}}$ ⟶ Divergent because $p = 2/3 \le 1$

$\sum_{n=1}^{\infty} \frac{1}{n}$ ⟶ Divergent because $p = 1 \le 1$

$\sum_{n=1}^{\infty} \frac{1}{n^2}$ ⟶ Convergent because $p = 2 > 1$

- **The Comparison Test**

Test 1: The comparison

If the behavior of the series $\sum_{n=1}^{\infty} b_n$ is known (convergent or divergent) then for the Series $\sum_{n=1}^{\infty} a_n$ we can say that it has the same behavior if the following conditionis satisfied

$$\lim_{n\to\infty} \frac{a_n}{b_n} = c \neq 0, \infty$$

Example 4

By using the comparison test study the following series

$$\sum_{n=1}^{\infty}\frac{7n-4}{n^3+2n-5}$$

Solution

We are going to compare the series

$$\sum_{n=1}^{\infty}\frac{7n-4}{n^3+2n-5} \sim \frac{n}{n^3} \sim \frac{1}{n^2} \quad \text{with} \quad \sum_{n=1}^{\infty}\frac{1}{n^2}$$

$$\lim_{n\to\infty}\frac{\left(\frac{7n-4}{n^3+2n-5}\right)}{\left(\frac{1}{n^2}\right)}=\lim_{n\to\infty}\frac{7n^3-4n^2}{n^3+2n-5}=7 \quad \neq 0,\infty$$

$p=2>1$

Then the series is convergent

Example 5

By using the comparison test study the following series $\sum_{n=1}^{\infty}\frac{5n^3-4}{n^{7/2}+2n-5}$

Solution

We are going to compare the series

$$\sum_{n=1}^{\infty}\frac{5n^3-4}{n^{7/2}+2n-5} \sim \frac{n^3}{n^{7/2}} \sim \frac{1}{n^{1/2}} \quad \text{with} \quad \sum_{n=1}^{\infty}\frac{1}{n^{1/2}}$$

$$\lim_{n\to\infty}\frac{\left(\frac{5n^3-4}{n^{7/2}+2n-5}\right)}{\left(\frac{1}{n^{1/2}}\right)}=\lim_{n\to\infty}\frac{5n^{7/2}-4n^{1/2}}{n^{7/2}+2n-5}=5 \quad \neq 0,\infty$$

$p=1/2\leq 1$

Then the series is divergent

- **The Ratio Test**

Test 2: The ratio test

For the series $\sum_{n=1}^{\infty} x_n$ we have

$$\lim_{n\to\infty}\left|\frac{x_{n+1}}{x_n}\right| = \begin{cases} > 1 & \text{the series is divergent} \\ < 1 & \text{the series is convergent} \\ = 1 & \text{the test fails} \end{cases}$$

Example 6

By using the ratio test study the following series $\sum_{n=1}^{\infty} \frac{1}{n\,3^n}$

Solution

We have $x_n = \frac{1}{n\,3^n}$, $x_{n+1} = \frac{1}{(n+1)\,3^{n+1}}$

then

$$\lim_{n\to\infty}\left|\frac{x_{n+1}}{x_n}\right| = \lim_{n\to\infty}\frac{\left(\frac{1}{(n+1)\,3^{n+1}}\right)}{\left(\frac{1}{n\,3^n}\right)} = \lim_{n\to\infty}\frac{n\,3^n}{(n+1)\,3^{n+1}} = \lim_{n\to\infty}\frac{n}{(n+1)3} = \frac{1}{3} < 1$$

Then the series is convergent

Example 7

By using the ratio test study the following series $\sum_{n=1}^{\infty} \frac{n!}{5^n}$

Solution

We have $x_n = \frac{n!}{5^n}$, $x_{n+1} = \frac{(n+1)!}{5^{n+1}}$

then

$$\lim_{n\to\infty}\left|\frac{x_{n+1}}{x_n}\right| = \lim_{n\to\infty}\frac{\left(\frac{(n+1)!}{5^{n+1}}\right)}{\left(\frac{n!}{5^n}\right)} = \lim_{n\to\infty}\frac{5^n(n+1)!}{5^{n+1}n!} = \lim_{n\to\infty}\frac{(n+1)n!}{5n!}$$

$$= \lim_{n\to\infty}\frac{(n+1)}{5} = \infty > 1$$

Then the series is divergent

- **The Integration Test**

Test 3: The integration test

For the positive series $\sum_{n=a}^{\infty} x_n$ by replacing each n with x to get f(x) such that this function is continuous, then

$\int_a^{\infty} f(x)\,dx =$ → $= \infty$ the series is divergent

→ $\neq \infty$ the series is convergent

Example 8

By using the integration test study the following series $\sum_{n=2}^{\infty}\frac{1}{n\,Ln\,n}$

Solution

We have $f(x)=\frac{1}{x\,Ln\,x}$

then

$$\int_2^{\infty}\frac{1}{x\,Lnx}dx=\int_2^{\infty}\frac{(1/x)}{Lnx}dx=Ln\left(Ln\,x\right)\Big|_{x=2}^{x=\infty}=Ln\left(Ln\infty\right)-Ln\left(Ln\,2\right)=\infty$$

Then the series is divergent

Example

By using the integration test study the following series $\sum_{n=2}^{\infty}\frac{1}{n\left(Ln\,n\right)^3}$

Solution

We have $f(x)=\frac{1}{x\left(Ln\,x\right)^3}$

then

$$\int_2^{\infty}\frac{1}{x\left(Lnx\right)^3}dx=\int_2^{\infty}\frac{1}{x}\left(Lnx\right)^{-3}dx=\frac{\left(Ln\,x\right)^{-2}}{-2}\Bigg|_{x=2}^{x=\infty}=\frac{\left(Ln\infty\right)^{-2}}{-2}-\frac{\left(Ln\,2\right)^{-2}}{-2}$$

$$=0+\frac{\left(Ln2\right)^{-2}}{2}\neq\infty$$

Then the series is convergent

Sheet (4)

(1) Find the following $\sum_{n=1}^{\infty} 3(\frac{1}{4})^{n-1}$

(2) Find the following $\sum_{n=1}^{\infty} 7(\frac{3}{5})^{n-1}$

(3) A rubber ball is dropped from a height of 30 meters. If it rebounds one-half the distance after each fall, use the geometric series to approximate the total distance the ball travels before coming to rest.

(4) Calculate the limit of the nth term of the following series and then discuss its convergence

(a) $\sum_{n=1}^{\infty} \frac{4n^2+2n+3}{3n^2+4}$

(b) $\sum_{n=1}^{\infty} \frac{1}{(1/2)^n+5}$

(c) $\sum_{n=1}^{\infty} (1+\frac{1}{n})^{3n}$

(d) $\sum_{n=1}^{\infty} (1+\frac{4}{n})^{2n+6}$

(5) Discuss the convergence of the following by using the comparison test

(a) $\sum_{n=1}^{\infty} \frac{1}{4n^2-1}$

(b) $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}\,(2n+5)}$

(c) $\sum_{n=1}^{\infty} \frac{n+3}{n^4+5n^2+4n+5}$

(d) $\sum_{n=1}^{\infty} \frac{\sqrt{n}}{n+4}$

(e) $\sum_{n=1}^{\infty} \frac{n^2+3}{5n^3-1}$

(f) $\sum_{n=1}^{\infty} \frac{n^5+4n^3+1}{2n^8+n^4+2}$

6) Discuss the convergence of the following by using the ratio test

(a) $\sum_{n=1}^{\infty} \frac{n}{3^n}$

(b) $\sum_{n=1}^{\infty} \frac{1}{n!}$

(c) $\sum_{n=1}^{\infty} \frac{1}{n\,2^n}$

(d) $\sum_{n=1}^{\infty} \frac{n!}{4^{n+3}}$

7) Discuss the convergence of the following by using the integration test

(a) $\sum_{n=3}^{\infty} \frac{1}{2n\,Ln\,n}$

(b) $\sum_{n=2}^{\infty} \frac{1}{n(Ln\,n)^5}$

(c) $\sum_{n=2}^{\infty} n^2 e^{-n^3}$

(d) $\sum_{n=2}^{\infty} \frac{1}{n^2} e^{\frac{1}{n}}$

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- Braun. M. Differential Equations and Their Application, Springer-Verlage , New York,2002.

Some laws

$$\frac{d}{dx}C=0 \qquad \frac{d}{dx}x^n=nx^{n-1} \qquad \frac{d}{dx}(fg)=f'g+fg' \qquad \frac{d}{dx}\left(\frac{f}{g}\right)=\frac{f'g-fg'}{g^2}$$

$$\frac{d}{dx}e^x=e^x \qquad \frac{d}{dx}Ln\,x=\frac{1}{x}$$

$$\frac{d}{dx}a^x=a^x\,Ln\,a \qquad \frac{d}{dx}Log_a x=\frac{1}{x}\frac{1}{Ln\,a}$$

$$\frac{d}{dx}\sin x=\cos x \qquad \frac{d}{dx}\cos x=-\sin x$$

$$\frac{d}{dx}\tan x=\sec^2 x \qquad \frac{d}{dx}\cot x=-\csc^2 x$$

$$\frac{d}{dx}\sec x=\sec x\tan x \qquad \frac{d}{dx}\csc x=-\csc x\cot x$$

$$\frac{d}{dx}\left(\sin^{-1}x\right)=\frac{1}{\sqrt{1-x^2}} \qquad \frac{d}{dx}\left(\cos^{-1}x\right)=\frac{-1}{\sqrt{1-x^2}} \qquad \frac{d}{dx}\left(\tan^{-1}x\right)=\frac{1}{1+x^2}$$

$$\frac{d}{dx}\left(\sec^{-1}x\right)=\frac{1}{x\sqrt{x^2-1}} \qquad \frac{d}{dx}\left(\csc^{-1}x\right)=\frac{-1}{x\sqrt{x^2-1}} \qquad \frac{d}{dx}\left(\cot^{-1}x\right)=\frac{-1}{1+x^2}$$

$$\cosh x=\frac{e^x+e^{-x}}{2} \qquad \sinh x=\frac{e^x-e^{-x}}{2} \qquad \tanh x=\frac{e^x-e^{-x}}{e^x+e^{-x}} \qquad (\cosh x)^2-(\sinh x)^2=1$$

$$\frac{d}{dx}\sinh x=\cosh x \qquad \frac{d}{dx}\cosh x=\sinh x$$

$$\frac{d}{dx}\tanh x=\sec h^2 x \qquad \frac{d}{dx}\coth x=-\csc h^2 x$$

$$\frac{d}{dx}\sec hx=-\sec hx\tanh x \qquad \frac{d}{dx}\csc hx=-\csc hx\coth x$$

$$\sinh^{-1}x=\ln\left(x+\sqrt{x^2+1}\right)\quad \cosh^{-1}x=\ln\left(x+\sqrt{x^2-1}\right)\quad \tanh^{-1}x=\frac{1}{2}\ln\left(\frac{1+x}{1-x}\right)$$

$$\operatorname{csch}^{-1}x=\ln\left(\frac{1+\sqrt{x^2+1}}{x}\right)\quad \operatorname{sech}^{-1}x=\ln\frac{1+\sqrt{1-x^2}}{x}\quad \coth^{-1}x=\frac{1}{2}\ln\left(\frac{x+1}{x-1}\right)$$

$$\frac{d}{dx}\left(\sinh^{-1}x\right)=\frac{1}{\sqrt{x^2+1}}\quad \frac{d}{dx}\left(\cosh^{-1}x\right)=\frac{1}{\sqrt{x^2-1}}\quad \frac{d}{dx}\left(\tanh^{-1}x\right)=\frac{1}{1-x^2}$$

$$\frac{d}{dx}\left(\operatorname{sech}^{-1}x\right)=\frac{-1}{x\sqrt{1-x^2}}\quad \frac{d}{dx}\left(\operatorname{csch}^{-1}x\right)=\frac{-1}{x\sqrt{1+x^2}}\quad \frac{d}{dx}\left(\coth^{-1}x\right)=\frac{-1}{x^2-1}$$

$$\int\frac{1}{\sqrt{a^2-x^2}}dx=\sin^{-1}\frac{x}{a}+C_1=-\cos^{-1}\frac{x}{a}+C_2\qquad \int\frac{1}{a^2+x^2}dx=\frac{1}{a}\tan^{-1}\frac{x}{a}+C_1=\frac{-1}{a}\cot^{-1}\frac{x}{a}+C_2$$

$$\int\frac{1}{x\sqrt{x^2-a^2}}dx=\frac{1}{a}\sec^{-1}\frac{x}{a}+C_1=\frac{-1}{a}\csc^{-1}\frac{x}{a}+C_2\qquad \int\frac{1}{\sqrt{x^2+a^2}}dx=\sinh^{-1}\frac{x}{a}+C$$

$$\int\frac{1}{\sqrt{x^2-a^2}}dx=\cosh^{-1}\frac{x}{a}+C\qquad \int\frac{1}{a^2-x^2}dx=\frac{1}{a}\tanh^{-1}\frac{x}{a}+C\qquad \int\frac{1}{x\sqrt{a^2-x^2}}dx=-\frac{1}{a}\operatorname{sech}^{-1}\frac{x}{a}+C$$

$$\int\sin x\,dx=-\cos x+C\quad \int\csc x\,dx=\ln\left|\csc x-\cot x\right|+C\quad \int\cos x\,dx=\sin x+C\quad \int\sec x\,dx=\ln\left|\sec x+\tan x\right|+C$$

$$\int\tan x\,dx=\ln\left|\sec x\right|+C\quad \int\cot x\,dx=\ln\left|\sin x\right|+C\quad \int\sinh x\,dx=\cosh x+C\quad \int\operatorname{csch}x\,dx=\ln\left|\tanh\frac{1}{2}x\right|+C$$

$$\int\cosh x\,dx=\sinh x+C\quad \int\operatorname{sech}x\,dx=\tan^{-1}(\sinh x)+C\quad \int\tanh x\,dx=\ln\left(\cosh x\right)+C\quad \int\coth x\,dx=\ln\left|\sinh x\right|+C$$

$$\sin x\ \cos y = 1/2\left[\sin(x+y)+\sin(x-y)\right] \qquad \cos x\ \sin y = 1/2\left[\sin(x+y)-\sin(x-y)\right]$$

$$\cos x\ \cos y = 1/2\left[\cos(x+y)+\cos(x-y)\right] \qquad \sin x\ \sin y = 1/2\left[\cos(x-y)-\cos(x+y)\right]$$

$$\sin(x-y) = \sin x\cos y - \cos x\sin y \qquad \sin(x+y) = \sin x\cos y + \cos x\sin y$$

$$\cos(x-y) = \cos x\cos y + \sin x\sin y \qquad \cos(x+y) = \cos x\cos y - \sin x\sin y$$

$$\tan(x-y) = (\tan x - \tan y)/(1+\tan x\tan y) \qquad \tan(x+y) = (\tan x + \tan y)/(1-\tan x\tan y)$$

$$\cot(x-y) = (\cot x\cot y + 1)/(\cot y - \cot x) \qquad \cot(x+y) = (\cot x\cot y - 1)/(\cot y + \cot x)$$

$$\sin 2x = 2\sin x\cos x \qquad -\sin x = \cos(x+\pi/2)$$

$$\cos 2x = \cos^2 x - \sin^2 x = 1 - 2\sin^2 x = 2\cos^2 x - 1 \qquad \cos x = \sin(x+\pi/2)$$

$$\tan 2x = \frac{2\tan x}{1-\tan^2 x}$$

$$Ln(xy) = Ln\,x + Ln\,y \qquad (x\pm y)^2 = x^2 \pm 2xy + y^2$$

$$Ln(x/y) = Ln\,x - Ln\,y \qquad x^2 - y^2 = (x-y)(x+y)$$

$$Ln\,x^n = n\,Ln\,x \qquad x^3 \pm y^3 = (x\pm y)(x^2 \mp xy + y^2)$$

$$az^2 + bz + c = 0\ ,\ z = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$$

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